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Superfluidity of Bose–Einstein condensates in lattices, modulated potentials and two dimensions

What the observables certify

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Abstract. A Bose–Einstein condensate in an optical lattice, a quasi-periodic or speckle potential, or a quasi-two-dimensional trap is superfluid only in part, and the superfluid part is never imaged directly. It is inferred from interference patterns after release, from Bragg and modulation spectra, from sound, from the response to a moving obstacle or to a phase twist, and from the decay of coherence with distance. This literature map collects the reference measurements on bosonic gases and sorts them by what each observable certifies. Stirring gives critical velocities between 0.07 and 0.6 of the sound speed, because vortices nucleate before Landau's threshold is reached, and in a lattice a dynamical instability competes with Landau's. Interference peaks survive above the critical temperature and in the Mott insulator, so they certify phase coherence, not superfluidity; a Talbot coherence length read with site-resolved images, which the authors attribute to the Bose-glass phase, is the one coherence measurement that goes further. The superfluid fraction comes from responses: sound across a modulation or in a triangular lattice, second sound at the Berezinskii–Kosterlitz–Thouless transition, Josephson dynamics and, in a calculation, winding numbers. Leggett's bound explains why a density modulation suppresses the superfluid fraction. The consequences are drawn for the phase diagram of the twisted-bilayer optical lattice, which rests on peak visibilities. Every number traces to a source checked for this map, in full or in part.

Keywords: Bose–Einstein condensate; superfluid fraction; Landau criterion; vortex nucleation; Leggett bound; Bogoliubov spectrum; Bragg spectroscopy; two-dimensional Bose gas; Berezinskii–Kosterlitz–Thouless transition; Bose–Hubbard model; Bose glass; time of flight; twisted-bilayer optical lattice

At a glance

10	28	8	200	200	22
chapters	pages of text	figures	sources	checked, 69 in full	open questions

Status: complete draft, first edition. Literature status 23 September 2026; checks of sources, numbers and quotations carried out on 23 September 2026 with AI assistance; see "How this map was made" on page i.

The problem

A Bose–Einstein condensate in a periodic, modulated or disordered potential is superfluid only in part, and that part has to be inferred from what can be measured: the interference and momentum patterns after release, the spectrum of excitations, the response to a stirrer or to a phase twist, and the decay of coherence with distance. The question is how the superfluid fraction, the Bogoliubov spectrum and the coherence of a weakly interacting condensate change when the potential removes translational invariance, by a lattice, by quasi-periodic or speckle disorder, or by squeezing the gas into two dimensions, and what each observable certifies about the phase of the gas.

It matters because these experiments, by the groups of Table 5.1, are the reference measurements against which any theory of a modulated Bose gas, including the twisted-bilayer one [1], must be tested. It is hard because superfluidity has several non-equivalent definitions (Landau's critical velocity, the non-classical response to a moving frame, phase stiffness, Leggett's bound), because a two-dimensional gas has no true condensate and its coherence is algebraic (Berezinskii–Kosterlitz–Thouless, BKT), because localised single-particle states in disorder or in narrow bands coexist with delocalised atoms, and because a time-of-flight image reflects the in-trap momentum distribution only in limits [2].

Established: Bogoliubov theory and its Bragg-spectroscopy confirmation [3, 4]; Landau's critical velocity measured by stirring [5, 6]; BKT in trapped and uniform 2D gases [7, 8]; Anderson localisation of expanding condensates in speckle and bichromatic potentials [9, 10]; Leggett's bound saturated in modulated gases and lattices (in the lattice, via BdG theory) [11, 12]; a Bose glass observed in an optical quasicrystal [13]. Not yet found in the literature checked for this map: a single account of which observable certifies which phase in modulated and disordered quasi-2D condensates, and of how localised excitations and band structure show up in interference and spectroscopy. Progress would be a checked dictionary from observables to phases, anchored in the measured numbers.

How this map was made: scope, method and conventions

Scope. Bosonic condensates in optical lattices, in quasi-periodic and speckle potentials and in quasi-two-dimensional geometry, and the observables that probe them (images, coherence, spectroscopy, stirring, sound, superfluid fraction). Supersolids enter only through one superfluid-fraction method; fermions and electronic systems are out of scope. Literature cutoff 23 September 2026; preprints are marked as such, and published versions are cited where they were found.

How it was compiled. The starting documents were the project's earlier map on twisted-bilayer optical lattices (161 checked sources, of which 85 are reused here) and its briefing. Three research agents extended them on critical velocity, Bragg spectroscopy, 2D gases, lattice superfluids, disorder and interference; four more read full texts on vortex nucleation and lattice instabilities, BKT and sound, Bose–Hubbard spectroscopy and the Bose glass. We looked up every source in a primary record (arXiv, publisher page, group copy) with AI assistance, in nine checking rounds and nine more for the second research round, including verbatim re-reads of the full text. Of the 200 sources the map cites, 69 were confirmed or corrected, 131 partly verified and 0 not verifiable.

Numbers. Each number and quotation traces to a line of the project's facts ledger, and no number was recomputed for this map, except two speed ratios in Fig. 2.2a and two inverted critical couplings, each marked. Quotations are verbatim from the checked source; numbers confirmed only in part carry "as reported" or "as quoted".

Conventions. $E_{\mathrm{R}} = \hbar^2 k^2 / 2m$ is the recoil energy of the lattice named in each case (sources write E_R, E_r or E_{rec}); J nearest-neighbour hopping, U on-site interaction; $\tilde{g} = \sqrt{8\pi}\, a_s / a_z$ the two-dimensional coupling, with a_s the scattering length and a_z the oscillator length of the tight axis; $D = n\lambda^2$ the phase-space density; f_{s} the superfluid fraction; c_{s} the sound speed; v_{c} the critical velocity.

In one page: the field and the argument

Cold-atom experiments have measured most of what the textbook says a superfluid does, in gases whose density can be imaged and whose potential is written with light. What they rarely measure is the superfluid fraction itself, and many phase diagrams of lattice and disordered gases rest on interference patterns instead. We argue in four steps that this matters.

1 **Superflow breaks down below the sound speed.** Stirring gives critical velocities between 0.07 and 0.6 c_s: $v_c \approx 1.6\,\mathrm{mm/s}$ against $c_s = 6.2\,\mathrm{mm/s}$ at the peak density of a three-dimensional condensate [5], $(0.07 \pm 0.02)\,c_s$ at low density [14] and $v_c/c_s = 0.3$–0.6 in two dimensions [6]. Vortex nucleation sets these values. Gross–Pitaevskii theory gives $(0.45 \pm 0.01)\,c$ for an impenetrable cylinder in 2D [15], and Kwon *et al.* measure "about 0.4" against the $0.37\,c_s$ of the large-cylinder theory they cite [16]. In a lattice, flow can also fail by a dynamical instability, whose region lies inside the Landau region [17]; one experiment characterised both [18].

2 **Interference measures coherence, not superfluidity.** In quantum Monte Carlo (QMC) "sharp peaks are not a reliable diagnostic of superfluidity": they survive above T_c, where superfluid density and condensate fraction vanish [19]. A quasi-2D gas can be bimodal and "quasi-condensed but is not superfluid" [20]. For coherent ^{87}Rb clouds the far field needs about 100 ms of flight, against the 20 ms typically used [2]. A Talbot coherence length that falls from 15(3) to 1.4(3) lattice sites, read together with site-resolved images, goes further: the authors attribute it to the Bose-glass phase [21].

3 **Superfluidity needs a response, and Leggett's bound explains its suppression.** Sound along and across a one-dimensional modulation gives $f_{s,x} = c_x^2/c_y^2$, and for a weakly interacting condensate Leggett's bound becomes an identity [11]. In a triangular lattice the dynamic method gives $f_s^{(yy)} = 0.82(2)$; the static method agrees [22]. In a uniform 2D gas the two sound speeds give a superfluid phase-space density that jumps "from 0 to 4 at the critical point" [8]. In a calculation the response is a winding number, which gives $f_s = 0$ in a quasicrystal Bose glass "except close to the separation line" [23]. Near the Mott insulator a lattice modulation reveals an amplitude (Higgs) mode [24], which tests the order parameter's amplitude, not the stiffness. The bound, $f_s \le 1/(\langle n\rangle\langle 1/n\rangle)$ with averages along the modulation, uses the density alone [25, 26]: flow must cross every low-density region, so a density that nearly vanishes somewhere forces f_s down, whatever the band structure.

4 **The twisted-bilayer diagram needs a response measurement.** The one twisted-bilayer experiment found for this map labels its phases by time-of-flight peaks at the primary and the moiré momenta; SF-II keeps only the primary ones and is called "superfluid only with short-range coherence" [1]. By steps 2 and 3, these labels certify coherence at two length scales. How much of the gas flows is not known: no superfluid-fraction or sound measurement on a moiré gas was found in the literature checked for this map. The in-situ images resolve the 4.35 μm moiré density [1]; in our reading, Leggett's upper bracket could be evaluated from such images, for the imaged species only.

What a calculation should report

Alongside the momentum distribution, a theory of a modulated condensate should give c_x and c_y, the f_s tensor from a phase twist and Leggett's brackets from its own density [11, 22, 26]. Chapter 3 explains how each is measured; chapter 6 argues the case.

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Terms from superfluidity, two-dimensional gases, optical lattices, disorder and cold-atom measurement, as this map uses them.

How to read this map

Twenty minutes. The problem, the one-page summary, then the key-points boxes at the end of each chapter.
Coming from theory, new to the lab. Chapter 3, the experimental toolbox and its dictionary.
Working on the twisted-bilayer problem. Chapters 6 and 8.
Looking for a research question. Chapter 9.
Boxes: key points, what a result means for the reader, open questions.

1 The topic in brief

What "superfluid" means in a cold-atom laboratory, the three ways a potential removes translational invariance, the experiments that define the field, and where the question of observables comes from.

A Bose–Einstein condensate (BEC) of a dilute gas is the simplest superfluid: "At zero temperature, a Galilean-invariant Bose fluid is expected to be fully superfluid" [11]. In the laboratory superfluidity is never seen directly; we infer it from a response. A stirrer or a moving lattice dissipates only above a critical velocity, part of the gas does not follow a slow motion of the potential, and sound travels at different speeds along and across a modulation. Phase coherence is a different quantity: it shows up as interference after the gas is released. The three responses probe the definitions of chapter 2, interference probes coherence, and the distinction between response and coherence runs through the whole map.

1.1 Three ways of removing translational invariance

A periodic optical lattice, made by standing waves of far-detuned light, turns the condensate into a Bloch state with a heavier band mass; deep lattices map onto the Bose–Hubbard model and its Mott insulator [27–29]. Disorder, from laser speckle, from two incommensurate lattices or from an optical quasicrystal, localises single-particle states and, with interactions, gives the Bose glass [9, 10, 13, 28, 30]. Squeezing the gas into a plane with $\hbar\omega_z \gg k_B T$, gn removes invariance along z and lets phase fluctuations destroy long-range order in the plane; the transition becomes the Berezinskii–Kosterlitz–Thouless (BKT) transition [31–34]. Experiments combine the three: square lattices on two-dimensional gases [35], a triangular lattice on a uniform plane [22], an eight-fold quasicrystal on a quasi-two-dimensional gas [13], and two twisted square lattices on a pancake condensate [1]. Figure 1.1 crosses the potential types with the observables, and Table 1.1 lists the platforms.

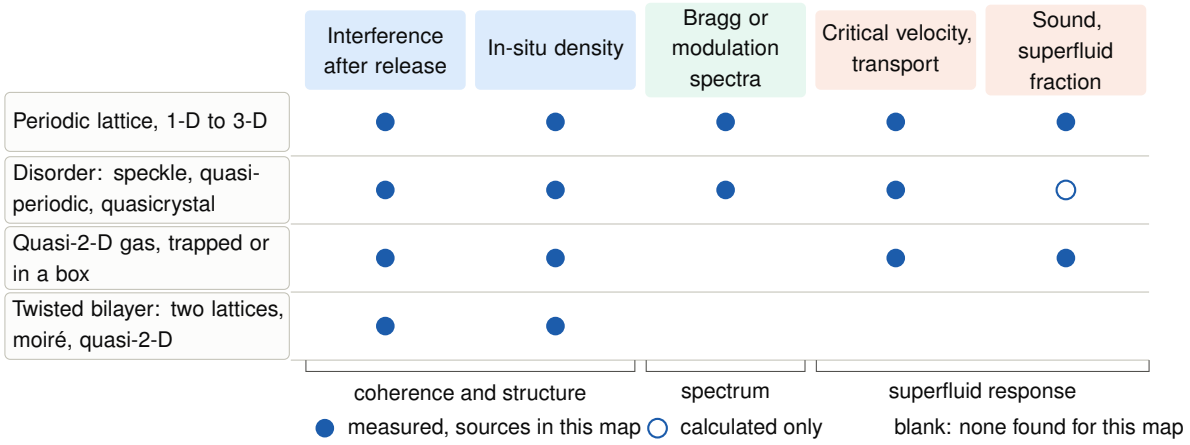


Figure 1.1: Potential types against observables. A filled dot means that at least one experiment among the sources of this map measured the quantity for that potential (Table 1.1 and chapter 5); an open dot means only calculations were found (the superfluid fraction of a Bose glass, from quantum Monte Carlo winding numbers [23]); a blank means none was found in the literature checked for this map, which is not proof of absence. The twisted-bilayer experiment also used microwave spectroscopy of the single-particle bands [1], which is not an excitation spectrum of the gas and is not marked.

1.2 The experiments that define the field

The critical velocity came first, as a test of Landau's criterion. A blue-detuned beam moved through a three-dimensional sodium condensate caused "Strong heating … only above a

Table 1.1: Platforms for bosons in modulated potentials. "Dim." is the dimension of the gas or of the modulation as the sources state it. Keys are a selection; chapter 5 gives the rest.

Platform	Potential	Dim.	What is measured	Sources
Stirred BEC, obstacle flow	moving blue-detuned beam in a trapped or oblate condensate; ring with a barrier	3-D, oblate	critical velocity from heating, drag or vortex nucleation; vortex dipoles and streets	[5, 14, 16, 36–38]
Bragg-probed BEC	harmonic trap; two Bragg beams	3-D	Bogoliubov spectrum; structure factor	[3, 4, 39, 40]
Trapped quasi-2-D gas	light sheet or planes of a 1-D lattice	2-D	interference of planes; equation of state; critical velocity; coherence exponent	[6, 7, 41, 42]
Uniform 2-D gas	optical box from a mask; single plane	2-D	sound; superfluid-density jump; defects after a quench	[8, 43, 44]
BEC in a 1-D lattice	shallow standing wave along one axis, static or moving	1-D lattice	effective mass; Josephson oscillations; energetic and dynamical instability; Bogoliubov bands	[18, 45–48]
Deep lattices, SF–MI	cubic or square lattice; arrays of tubes	3-D, 2-D, 1-D	interference visibility; Mott gap; in-situ shells; amplitude mode	[24, 29, 35, 49, 50]
Modulated gas, f_s	1-D modulation, 1-D lattice or triangular lattice	2-D gas; 1-D lattice	superfluid fraction from sound, Bragg phonons or the density profile	[11, 12, 22]
Disordered gas	laser speckle; bichromatic lattice	1-D, 3-D	localised expansion; mobility edge; interaction-driven delocalisation; transport	[9, 10, 51–54]
Optical quasicrystal	four 1-D lattices at 45°, eight-fold	2-D	coherence of the diffraction pattern; localisation depth; Bose glass	[13, 55–57]
Twisted-bilayer optical lattice	two square lattices rotated by 5.21°, one per spin state; microwave coupling	quasi-2-D	in-situ moiré density; visibilities of primary and moiré peaks; band spectroscopy	[1]

critical velocity" [5], and the drag on the beam set in near a tenth of the sound speed [14]. Later experiments imaged what the obstacle leaves behind: single vortex dipoles in an oblate condensate [36], a critical velocity that depends on the height of the obstacle [16], and a von Kármán vortex street [37]. The threshold appears in other geometries too. A persistent current in a ring stopped abruptly when the flow at a barrier exceeded a critical value [38]. A two-dimensional gas flowed past a micrometre-sized obstacle without friction only where it was degenerate enough [6], and a moving lattice separated energetic from dynamical instability [18, 47]. Bragg spectroscopy, for its part, measured the Bogoliubov spectrum, its phonon part and the Bogoliubov transformation itself [3, 4, 39, 40]; Ozeri *et al.* review this work [58].

In two dimensions Hadzibabic *et al.* observed the BKT crossover in a trapped ^{87}Rb gas: "the loss of long-range coherence coincides with the onset of proliferation of free vortices" [7]. Scale invariance and the equation of state followed [41, 59], and the critical point was measured against classical-field theory [60]. Uniform gases in optical boxes [43] then gave sound [44] and, from first and second sound, the superfluid density with its universal jump [8].

In lattices, a condensate in a one-dimensional standing wave showed a heavier effective mass, Josephson oscillations between wells and a velocity above which its motion dissipates [45, 46]. Greiner *et al.* deepened a three-dimensional lattice and saw "a transition … from a superfluid

to a Mott insulator phase" [29]. Modulation and Bragg spectroscopy then measured the Mott gap, the Bogoliubov bands and the amplitude mode of lattice superfluids [24, 49, 61–63], and quantum-gas microscopes imaged the Mott shells site by site [50, 64].

Disorder entered with Anderson localisation of expanding condensates in speckle and in bichromatic lattices [9, 10]. Experiments then turned to interacting regimes, with reported Bose-glass signatures [53, 54, 65, 66]. In an eight-fold optical quasicrystal Schneider's group observed "a Bose-glass-to-superfluid transition" by probing coherence [13, 55, 56], and a preprint follows quenches across that transition [57].

The measurements of the superfluid fraction itself found for this map start in 2023: from the anisotropy of sound in a two-dimensional gas with a one-dimensional modulation [11], in a one-dimensional lattice inferred as an anisotropic superfluid density from Bragg-measured sound speeds, f_s compared with Leggett via BdG theory [12], as a full tensor in a triangular lattice [22], and from Josephson oscillations in a dipolar supersolid [67]. The twisted-bilayer optical lattice closes Table 1.1: two square lattices, one per spin state of ^{87}Rb, rotated by 5.21° and coupled by a microwave, on a quasi-two-dimensional condensate [1].

1.3 Where the question of observables comes from

Most phase diagrams of lattice and disordered Bose gases are read from interference after release. In the quasicrystal, "A high-contrast interference pattern composed of sharp peaks indicates the presence of long-range phase coherence, the signature of the superfluid phase" [13], and the twisted-bilayer phases are labelled by which families of peaks are present [1]. Three results show that this reading needs care. Quantum Monte Carlo (QMC) calculations for the three-dimensional Bose–Hubbard model found that "sharp peaks are not a reliable diagnostic of superfluidity", because they survive above the critical temperature where the superfluid density and the condensate fraction vanish [19]. A quasi-two-dimensional gas can be "quasi-condensed but … not superfluid" [20]. And the far field that turns an image into a momentum distribution is reached only after about 100 ms of flight, against the usual 20 ms [2]. Interference measures coherence over some length; superfluidity is a response. Which observable certifies which phase is the question of this map: chapter 3 describes the measurements, chapter 4 the benchmarks, and chapter 6 states the position.

Key points

- Superfluidity is inferred from a response: critical velocity, the part of the gas that does not follow a moving potential, or sound; interference after release measures coherence.
- Critical-velocity experiments (stirring, vortex streets, moving lattices) are the oldest response test; they measure thresholds below the sound speed.
- Translational invariance is removed by a lattice, by disorder or quasi-periodicity, or by squeezing the gas into two dimensions; experiments combine them.
- The measurements of the superfluid fraction of modulated condensates found for this map date from 2023 on, by two groups using sound and one using Josephson oscillations.
- Sharp interference peaks survive above the critical temperature in QMC, so phase diagrams read from peak visibilities need a response measurement to be read as superfluid.

2 Foundations

The definitions behind the observables: Landau's criterion and why measured critical velocities fall below it, the superfluid fraction and Leggett's bound, the Bose–Hubbard model and the two-dimensional gas.

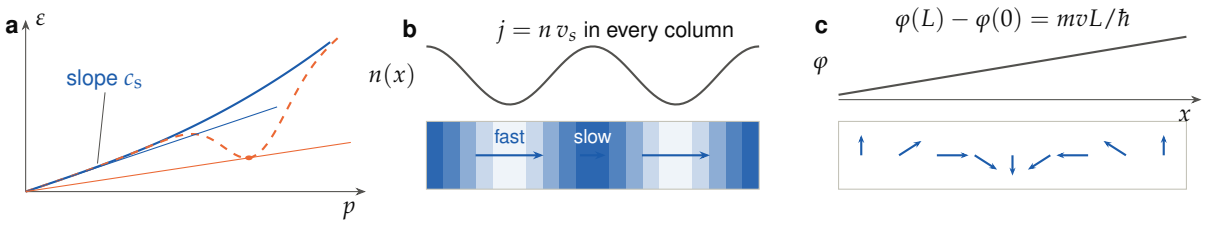


Figure 2.1: (a) Landau's criterion: $v_c = \min_p \varepsilon(p)/p$ is the slope of the tangent from the origin to the spectrum, c_s for Bogoliubov (solid), lower with a minimum (dashed). The tangent is an upper bound; measured critical velocities fall below it (Fig. 2.2a). (b) Leggett's columns (darker is denser): one current through all, fast where the density is low, so $f_s \le [\langle n\rangle\langle 1/n\rangle]^{-1}$. (c) A phase winding imposes a flow v at an energy cost $f_s N m v^2/2$.

2.1 The Bogoliubov spectrum and Landau's criterion

Linearising the Gross–Pitaevskii equation [68, 69] about a uniform condensate of density n gives $\varepsilon(p) = [c_s^2 p^2 + (p^2/2m)^2]^{1/2}$ with $mc_s^2 = gn$, g the contact coupling: phonons below the inverse healing length $1/\xi$, free particles above, and a quantum depletion [70]. Bragg spectroscopy confirmed it, with a mean-field shift of $(0.54 \pm 0.07)\, n_0 g/h$ against the predicted $4n_0 g/7h$ (written with U for g) [3] and the full spectrum giving an effective sound speed $c_{\mathrm{eff}} = 2.0 \pm 0.1$ mm/s [4]. In a lattice the spectrum splits into Bogoliubov bands [71].

A fluid flowing at velocity v past an obstacle can lower its energy by creating an excitation of momentum p only if $\varepsilon(p) - pv < 0$. Flow is therefore dissipationless below $v_c = \min_p \varepsilon(p)/p$, the slope of the tangent from the origin to the spectrum (Fig. 2.1a) [72], a necessary condition for metastable superflow in Dalibard's course [73]. For the Bogoliubov spectrum the tangent is the phonon line and $v_c = c_s$, in three dimensions and in two [32, 38]; the measured linear regime "provides an upper limit for the superfluid critical velocity, by the Landau criterion" [4]. A weak impurity feels a drag only above c_s in three dimensions, but in the exact one-dimensional theory "for an arbitrarily small velocity" [74]. Vortices and lattice instabilities lie outside the construction; they set the measured thresholds.

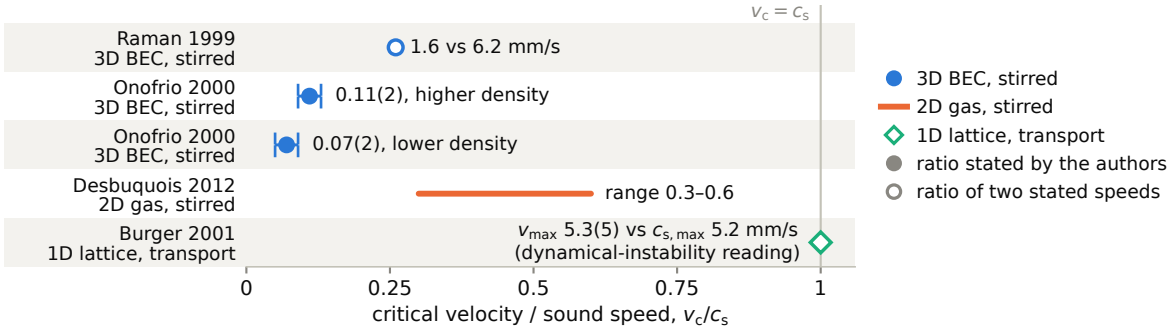
2.2 Why measured critical velocities are lower

Stirred condensates dissipate well below c_s (Fig. 2.2a). In the experiment of Raman *et al.* a blue-detuned beam scanned through a sodium condensate depleted it, with a crossover at about 1.6 mm/s and almost full depletion near 6 mm/s after 900 ms, against a sound speed of 6.2 mm/s at the peak density. Feynman's vortex-pair estimate gives 0.7 mm/s; the local-sound criterion of the nonlinear-Schrödinger obstacle problem [75], $v_c = \sqrt{2/11}\, c_{\max}$ as the authors quote it, gives 2.6 mm/s [5]. Onofrio *et al.* measured the drag, which "arises due to the pressure difference across the moving object": it set in at $v_c = (0.11 \pm 0.02)\, c_s$ and $(0.07 \pm 0.02)\, c_s$ at higher and lower density, and a ratio of drag coefficients of 1.9 ± 0.2 (predicted 2.1) tied it to vortex shedding [14].

Simulations name the mechanism. In two-dimensional flow past an impenetrable cylinder the drag vanishes below $v_c = (0.45 \pm 0.01)\, c$ and vortices are emitted above [15]; for a large cylinder, $R \gg \xi$, calculations quoted by Kwon *et al.* give $0.37\, c_s$ [16]. In a trapped three-dimensional

condensate "vortex lines first appear" at the edge, where the sound speed is lower, and an oscillating obstacle gives $v_c \simeq 0.20\,c_s$ or $0.13\,c_s$ depending on its height [76]. Dalibard's notes give the scale: vortex pairs form above $\hbar/(m\ell)$ for an obstacle of size ℓ, below c_s when $\ell \gg \xi$ [73]. Experiments then watched the vortices. In an oblate ^{87}Rb condensate a single vortex dipole appeared between 170 and 190 µm/s against $c \sim 1700$ µm/s, $v_c \sim 0.1c$, in excellent agreement with simulation [36]. In oblate sodium condensates v_c/c_s was "about 0.40" for the stiffest obstacle, with a dip at $V_0 \approx \mu$ [16], and a von Kármán vortex street formed above $v_c = 1.11(5)$ mm/s, with $c_s \approx 5.1$ mm/s and a Strouhal number near 0.2 [37]. In a two-dimensional ^{87}Rb gas stirred by a 2 µm beam, $v_c/c_s = 0.3$–0.6, and frictionless flow set in at a local $\mu/k_BT \approx 0.24$, above the 0.15 predicted for an infinite uniform system [6]; classical-field simulations trace the heating to vortex–antivortex pairs [77]. In a ring the persistent current decayed at $v_c/c_{\text{eff}} \approx 0.6$ for all barrier settings [38]. The ratios 0.07–0.6 do not contradict Landau: they are thresholds for nucleating vortices, set by the obstacle and the local density.

a Measured critical velocities



b Superfluid fraction: what was measured

experiment	method	potential	result as stated
Chauveau 2023	sound anisotropy c_x^2/c_y^2; Leggett formula on the density	2D ^{87}Rb, 1D modulation, $d = 3.93(4)$ µm, $V_0 = 0$–80 nK	Leggett bound exact for 1D modulation (confirmed); f_s plotted, no numeric value given
Tao 2023	Bragg-measured sound speeds (anisotropic)	1D lattice, $a = 266$ nm, depth up to 10 E_R	anisotropic f_s inferred from sound speeds; Leggett via BdG theory; no number
Christodoulou 2021	first and second sound	2D gas across the BKT transition	superfluid density consistent with BKT, including the universal jump at T_c
Biagioni 2024	Josephson oscillations of single supersolid cells	dipolar supersolid (self-modulated density)	"a relatively large sub-unity superfluid fraction"; no numeric value given
Rabec 2026	density + continuity eq. (static); compressibility + sound (dynamic)	2D ^{87}Rb, triangular lattice, $d = 6.0(1)$ µm, V_0 up to 4.7 μ_0	$f_s^{(yy)} = 0.82(2)$ (dynamic method) at $V_0 \simeq 2.4\,g\rho_0$; static method agrees; both agree with GPE and the Leggett bounds
Gautier 2021 (QMC)	path-integral QMC (winding numbers)	8-fold quasicrystalline lattice	Bose glass: $f_s \approx 0$; $V_c/E_R \simeq 1.76 \pm 0.01$ (weak interactions); smooth compressibility crossover

Values as stated by the cited authors; open marks are ratios of two stated speeds.

Figure 2.2: (a) Measured v_c/c_s against the Landau value 1 of a uniform condensate; the ratios depend on the obstacle and on where c_s is taken. Filled marks are ratios the authors state, open marks the ratio of two speeds they state. The lattice point, a ratio near 1, compares a maximal velocity with a computed sound speed and is not called a Landau threshold by its authors [45]; it admits a dynamical-instability reading [17]. (b) Superfluid-fraction measurements as stated; $\mu_0 = g\rho_0$ is the mean-field energy at the box density ρ_0; the last row is a calculation [23].

2.3 Moving lattices: energetic and dynamical instability

Wu and Niu separate an energetic (Landau) instability, where the moving Bloch state becomes a saddle point of the energy, from a dynamical one, where Bogoliubov frequencies turn complex and perturbations grow exponentially; the second region lies inside the first, and in tight binding

it starts at half the Brillouin zone [17], $|q| > \pi/(2a)$ in Dalibard's notes [73]. Menotti *et al.* add two density-dependent effective masses to the tight-binding picture [78]. An energetic instability needs something to take up the energy: De Sarlo *et al.* found an almost pure condensate insensitive to the lattice velocity after 15 s, whereas a cloud with about 35 % thermal fraction decayed even at small velocity, with $\tau_{\mathrm{EI}} = (416 \pm 55)$ ms: the thermal fraction triggers the energetic instability [18]. The dynamical instability needs no bath: the lifetime in a moving lattice fell from ≈10 s to ≈10 ms with "a discontinuity around $q = 0.55 q_B$" (q_B the quasimomentum at the zone edge), "unambiguously attributed to the onset of dynamical instability" after comparison with Gross–Pitaevskii theory [47], and De Sarlo *et al.* give the onset at $q = 0.56\, q_B$ for a depth $s = V_0/E_{\mathrm{R}} = 0.2$, as reported [18].

Other readings exist. Burger *et al.* measured a full loss of superfluidity at $v_{\max} = (5.3 \pm 0.5)$ mm/s against a computed $c_{s,\max} \simeq 5.2$ mm/s, and read the lattice as a rough medium in which $v_c(r)$ equals the local sound speed, without naming Landau [45]; Wu and Niu suggested that dynamical instability may play a role there [17]. In one-dimensional tubes damping appeared at $q_{\max} \approx \pi/5$ for $V < 2E_{\mathrm{R}}$, below the $\pi/2$ threshold, and was attributed to quantum fluctuations [79]. In three dimensions the critical momentum fell from 0.5 recoil momenta in a very weak lattice to zero at the Mott transition, where a fit of the predicted form $p_c \propto \sqrt{1 - u/u_c}$, with $u = U/J$, gave $u_c = 34.2(2.0)$ [80]. Which instability was crossed is decided by a model of each experiment.

2.4 The superfluid fraction and Leggett's bound

"The superfluid fraction f of a quantum fluid is defined in terms of the response of the system to a weak and constant drag" [26]: the part of the gas that a slowly moving potential does not drag along. For N atoms of mass m this reads, in momentum and in energy (the energy form as Pérez-Cruz *et al.* write it [26]),

$$f_{\alpha\beta} = 1 - \lim_{v\to 0} \frac{\langle \hat{P}_\alpha \rangle}{N m v_\beta}, \qquad E(v) = E(0) + f_{\mathrm{s}} \frac{N m v^2}{2} \quad (v \to 0), \tag{1}$$

a rank-2 tensor in an anisotropic potential [81]. Calculations use the phase twist (Fig. 2.1c), as a helicity modulus [82] or quantum Monte Carlo (QMC) winding numbers [83]. With c_x along and c_y across a one-dimensional modulation, Chauveau *et al.* write $c_x^2 = f_{s,x}/(m\kappa)$, κ the compressibility, hence $f_{s,x} = c_x^2/c_y^2$, and state that for their weakly interacting condensate $f_{s,x}$ coincides with m/m_x^*, m_x^* the band mass [11]; the lattice sound speed $1/\sqrt{\kappa m^*}$ carries it too [84, 85].

Leggett's bound follows from one picture. For a density $n(x)$ and a slow flow along x with a phase depending on x only, the current $j = n(x)\, v_s(x)$ is the same through every column (Fig. 2.1b), so the flow is fast where the gas is thin. The twist fixes $\langle v_s \rangle = j\langle 1/n \rangle = v$, the kinetic energy per length is $\frac{m}{2}\langle n v_s^2 \rangle = \frac{m}{2} v^2/\langle 1/n \rangle$, and equating it to $f_{\mathrm{s}} \langle n \rangle m v^2/2$ gives $1/(\langle n \rangle \langle 1/n \rangle)$. The true twisted ground state lies at or below this phase-only trial state, hence Leggett's bound $f_{\mathrm{s}} \le 1/(\langle n \rangle \langle 1/n \rangle)$ [25, 26]. "The bound reduces to an identity in the case of a weakly interacting BEC subject to a 1D periodic potential" [11]: a condensate is one orbital whose only freedom is the phase. Since $\langle n \rangle \langle 1/n \rangle \ge 1$, any modulation lowers f_{s}, most where the columns are thinnest. In two dimensions a density $n(x, y)$ gives two brackets for a twist along x [26],

$$f^- = \frac{1}{\bar{n}} \Big\langle \frac{1}{\langle 1/n \rangle_x} \Big\rangle_y \;\le\; f_{\mathrm{s}} \;\le\; f^+ = \frac{1}{\bar{n}} \frac{1}{\big\langle 1/\langle n \rangle_y \big\rangle_x} : \tag{2}$$

f^+ treats each column as a whole, f^- lets each line of constant y carry its own current. "The bracketing is tighter for a 2D optical potential than for a grid of columns" [26], and the triangular-

lattice measurement agrees "with the Leggett bounds" [22]. A preprint tests the bound for strong interactions (not checked for this map) [86]; another studies its angular dependence [87].

2.5 The Bose–Hubbard model

In a deep lattice the atoms occupy the lowest Wannier orbital w of each site, and [27, 28]

$$H = -J \sum_{\langle ij \rangle} b_i^\dagger b_j + \frac{U}{2} \sum_i n_i(n_i - 1) - \mu \sum_i n_i, \tag{3}$$

with both parameters set by the laser light [27]. The papers write U/t or U/J; we write J throughout. $J \approx (4/\sqrt{\pi})\, E_\mathrm{R}\, s^{3/4} e^{-2\sqrt{s}}$ falls exponentially with depth [88], while $U \propto g \int |w|^4$ grows slowly: a deeper lattice turns a superfluid (SF) into a Mott insulator (MI) with "exact numbers of atoms … localized at individual lattice sites" and "a gap in the excitation spectrum" [29].

At unit filling mean field gives $U/zJ \approx 5.83$, z the number of nearest neighbours [89]: $U/J \simeq 23.31$ on the square lattice [90]. For the cubic lattice Mun *et al.* and Mark *et al.* quote 34.8 (Mun: 5.8×6), which Mark *et al.* call "close to" the QMC result, both theory values [80, 91]. QMC gives $(U/J)_c = 29.34(2)$ in three dimensions [92] and $(J/U)_c = 0.05974(3)$ in two [93], and $U_c/J = 16.7424(5)$ in later QMC [94]; the density-matrix renormalisation group gives $J_c/U = 0.277 \pm 0.01$, that is $(U/J)_c \approx 3.61$ (our inverse of $J_c/U = 0.277(10)$), in one [95]. Early QMC [96] and two reviews [97, 98] cover the model.

SF and MI differ in interference visibility, although the visibility survives in part in the MI through particle–hole admixtures [29, 99, 100]. The MI also has a gap, seen as a discrete modulation spectrum with its first resonance near U [49], integer filling, imaged as shells site by site [50, 64], and squeezed number statistics: local fluctuations are suppressed [101], and so are spin-changing collisions, "direct evidence for the emergence of number-squeezed states" [102]. A lattice superfluid, in turn, is far from a pure condensate: depleted fractions "in excess of 50 %" were seen in three dimensions [103]. The model itself has limits. In hexagonal potentials at 15 E_R it describes only the first Mott lobe [104], Dutta *et al.* review the non-standard terms [105], and number-dependent shifts reveal multi-body interactions [91].

2.6 Quasi-two-dimensional gases and the BKT transition

In a quasi-two-dimensional gas [31], with $a_z = \sqrt{\hbar/m\omega_z} \ll \xi, \lambda/\sqrt{2\pi}$ (chapter 3), collisions stay three-dimensional since $a_z \gg a_s$ [32], and excited z levels lower T_c [106]. Interactions enter through $\tilde{g} = \sqrt{8\pi}\, a_s/a_z$; the review also gives the 2-D scattering length $a_2 = a_z\sqrt{\kappa}\, \exp(-\sqrt{\pi/2}\, a_z/a_s)$, with a constant $\kappa \approx 3.5$ (Eq. 76 of [32]), and Petrov *et al.* the log-corrected coupling [31]. Experiments have $\tilde{g} = 0.093$ [6], ≈0.1 [41], 0.05–0.26 [59], 0.16(1) [44], 0.158(4) [22] and 0.64(3) in ^{39}K [8]; $c_\mathrm{s} = \hbar\sqrt{\tilde{g}n}/m \approx 1.6\,\mathrm{mm/s}$ at $n = 50\,\mu\mathrm{m}^{-2}$ [6].

In an infinite plane at $T > 0$ first-order coherence decays algebraically [33], and superfluidity ends as vortex pairs unbind [34]: the Berezinskii–Kosterlitz–Thouless (BKT) transition. Classical-field Monte Carlo gives a critical phase-space density $D_c = \ln(380/\tilde{g})$, 380 ± 3, and a critical chemical potential $\mu_c \propto \ln(\xi_\mu/\tilde{g})$, $\xi_\mu = 13.2 \pm 0.4$ [107, 108]. The superfluid phase-space density $D_\mathrm{s} = n_\mathrm{s}\lambda^2$ jumps by 4 [32, 109], seen in first and second sound [8]. In a trap QMC gives $T_\mathrm{KT} \simeq 0.70\, T^0_\mathrm{BEC}$ and a superfluid "with vanishing condensate fraction" [110]; finite size gives a BEC–BKT crossover [111], and classical fields show coherence above T_BKT, exponent 0.25 at T_BKT near the centre, vortices mostly paired [112]. A bimodal gas can be "quasi-condensed but … not superfluid" [20].

Key points

- Landau gives $v_c = c_s$; measured v_c/c_s runs from 0.07 to 0.6 because obstacles nucleate vortices, and moving lattices add energetic and dynamical instabilities.
- $f_{s,x} = c_x^2/c_y^2$; Leggett's bound is exact for a weak condensate in a 1-D modulation.
- $(U/J)_c = 29.34(2)$, 16.7424(5), about 3.61 (our inverse) in 3-D, 2-D, 1-D, below mean field; BKT: $D_c = \ln(380/\tilde{g})$, $n_s\lambda^2 = 4$, $T_{\mathrm{KT}} \simeq 0.70\, T_{\mathrm{BEC}}^0$ in a trap.

3 Context: the experimental toolbox, for theorists

How a quasi-two-dimensional or lattice condensate is prepared, probed and imaged, and which theoretical quantity each measured signal stands for.

Every experiment in this map traps a gas, shapes its potential, probes it and takes an image (Fig. 3.1); the probe and the image fix which quantity is measured, with the pitfalls of Table 3.1.

3.1 Making a quasi-two-dimensional gas

A gas is quasi-two-dimensional when the tight confinement freezes the motion along z, which requires "the frequency of the tight confinement ω_0 to be much larger than the gas temperature T and the mean-field interparticle interaction $n_0 g$" [31]. Hadzibabic and Dalibard write the equivalent condition $a_z \ll \xi, \lambda/\sqrt{2\pi}$, with $a_z = \sqrt{\hbar/m\omega_z}$ the oscillator length, ξ the healing length and λ the thermal wavelength [32]. Collisions stay three-dimensional because $a_z \gg a_s$, and the scattering length a_s enters through $\tilde{g} = \sqrt{8\pi}\, a_s/a_z$, which "varies between $\sim 10^{-1}$ and $\sim 10^{-2}$" in experiments [32]: $\tilde{g} \approx 0.1$ at $\omega_z/2\pi = 1.9(2)$ kHz in the equation-of-state work of Yefsah *et al.* [41], $\tilde{g} = 0.158(4)$ at $\omega_z/2\pi = 4.1(2)$ kHz in the 2026 triangular-lattice experiment [22].

The tight confinement is made with light. A standing wave along z made of two running beams gives a stack of planes [32], and interfering two planes after release revealed the BKT crossover by matter-wave heterodyning [7]; a vertical double well with about 1 kHz per plane serves the same purpose [42]. An accordion lattice, with spacing tuned "between 11 and 2 μm", loads a single plane and compresses it to $\omega_z/2\pi > 10$ kHz [113, 114]. A beam with an intensity node at $z = 0$ gives $\nu_z = 350$–1500 Hz, and a mask on a second beam cuts a box of 200–900 μm^2 in the plane [43]; a digital micromirror device (DMD) imaged onto the atoms makes such potentials programmable [115]. A box removes the averaging over densities that a harmonic trap imposes, and the sound and superfluid-fraction experiments of Ville *et al.* and Rabec *et al.* use one [22, 44]. The review of Hadzibabic and Dalibard [32] and Perrin's lecture notes, posted as a preprint [116], cover this physics.

3.2 Imaging: in situ and after time of flight

An in-situ absorption image gives the column density, the density integrated along the line of sight. With the local density approximation (LDA), which treats each point of the trap as a uniform gas at the local chemical potential, one image yields an equation of state; in two dimensions the dimensionless quantities depend on μ/k_BT only [41]. In a lattice the same route gives a local compressibility, strongly suppressed in the Mott domains, and density fluctuations that act as a thermometer through the fluctuation–dissipation theorem [101]; in-situ density has also followed slow mass transport [117] and quantum-critical scaling with dynamical exponent

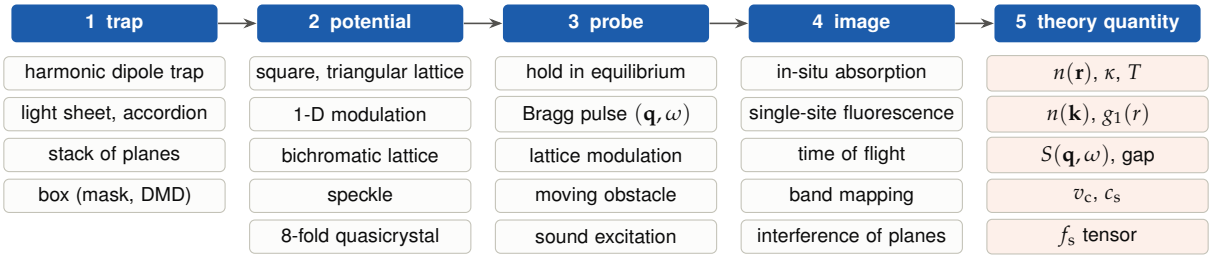


Figure 3.1: The experimental sequence as a theorist meets it. Each column lists the options used by the experiments of this chapter; the last is what the analysis returns. The choice of probe and image fixes which quantity is measured (Table 3.1).

$z = 2.0(3)$, $\nu = 0.54(5)$ [118]. Quantum-gas microscopes resolve single sites. In the microscope of Bakr *et al.* (lattice period 680 nm) atoms of the $n = 1$ shell are detected with probability 94.9 ± 0.7 % at 16 E_R, and shells up to $n = 4$ appear through parity projection: the microscope sees the occupation modulo two [50]. The microscope of Sherson *et al.* takes temperature and entropy from single images and shows "Mott-insulating plateaus with near-zero entropy" [64].

Most lattice experiments image after time of flight (TOF): the potential is switched off and the cloud expands for tens of milliseconds. Only in the far field does the image reproduce the in-trap momentum distribution. Gerbier *et al.* give the far-field time $t_{FF} = mR_0 l_c/\hbar$, with R_0 the cloud size and l_c the coherence length; for coherent ^{87}Rb samples $t_{FF} \approx 100$ ms against a typical 20 ms, so "the far-field condition is usually not met" [2]. In two dimensions a focusing technique gives momentum-space images directly [119]. Ramping the lattice down in 2 ms maps the first band onto the first Brillouin zone [120], and correlations of the density noise in the expanding cloud reveal the order of a Mott insulator [121]. Coherence itself is read from the interference of two condensates [122, 123], from the correlation function, which decays as a Gaussian above T_c and to a plateau below [124], from the visibility and width of lattice TOF peaks [99, 100, 125], and in low dimensions from the statistics of fringe contrast [126].

3.3 Spectroscopy and stirring

Bragg spectroscopy uses two beams with momentum difference $\hbar\mathbf{q}$ and frequency difference ω; the number of atoms transferred measures the dynamic structure factor $S(\mathbf{q}, \omega)$. In trapped condensates it gave the mean-field shift [3], the suppressed phonon response [39], the whole Bogoliubov spectrum (chapter 2) [4] and the Bogoliubov transformation itself [40]; Ozeri *et al.* review these measurements [58]. In lattices, momentum-resolved Bragg spectra show the band structure and the limits of Bogoliubov theory [48, 62, 71]; the resonance falls faster with depth than Bogoliubov theory predicts [63], and in one-dimensional gases the response "drastically drops" in the insulating regime [61].

Lattice-modulation spectroscopy shakes the lattice depth and measures the energy absorbed [127]. The superfluid shows a broad continuum, the Mott insulator a discrete spectrum with a first resonance near U and a second at (1.91 ± 0.04) times the first [49]; precision versions resolve number-dependent shifts from effective multi-body interactions [91]. In two dimensions the onset of absorption softens towards the critical point, $j_c \simeq 0.06$ from quantum Monte Carlo (QMC), with $j = J/U$: the amplitude ("Higgs") mode [24], which Bragg scattering also detects [128].

Stirring tests Landau's criterion, a necessary condition for the metastability of a superflow [73]. A blue-detuned beam moved through a sodium condensate heated it strongly only above about 1.6 mm/s [5]; in a two-dimensional ^{87}Rb gas a beam of waist 2.0(5) μm gave critical velocities of 0.5–1.0 mm/s [6]. Rings with a barrier [38] and moving lattices [45, 47, 80] probe the same

physics.

3.4 Disorder and quasi-periodic potentials

Speckle, laser light passed through a diffuser, is a disorder with a finite correlation length [30, 129]. In one dimension it localises expanding atoms exponentially only when their de Broglie wavelength exceeds "an effective mobility edge" set by the speckle cutoff [9]; in three dimensions localised and diffusive components separate [51] and the mobility edge has been measured [52]. A second, incommensurate lattice gives quasi-periodic disorder (830 and 1076 nm, as reported [65]), and the critical disorder for localising a non-interacting condensate scales with J [10]. Four one-dimensional lattices at $45(1)^\circ$ with $\lambda = 726\,\mathrm{nm}$ make an eightfold quasicrystal [55], where the two-dimensional Bose glass was observed [13, 56].

3.5 Measuring the superfluid fraction

Chauveau *et al.* imprint a one-dimensional modulation of period $d = 3.93(4)\,\mu$m and amplitude up to 80 nK on a two-dimensional ^{87}Rb gas, measure the sound speed along and across it, obtain $f_{s,x} = c_x^2/c_y^2$ and compare it with Leggett's formula on the imaged density [11]. Tao *et al.* measure sound speeds by Bragg scattering in a lattice of period $a = 266\,\mathrm{nm}$, up to $10\,E_\mathrm{R}$, and infer the anisotropic superfluid density from them; f_s is compared with Leggett via BdG theory [12]. Rabec *et al.* use a triangular lattice of period $d = 6.0(1)\,\mu$m in a box of 42(1) μm; the dynamic method (compressibility and sound speed) gives $f_s^{(yy)} = 0.82(2)$ at $V_0 \simeq 2.4\,g\rho_0$, ρ_0 the density in the box; the static method (density profile and continuity equation) agrees [22]. Biagioni *et al.* obtain f_s from the Josephson oscillations of single supersolid cells [67].

Table 3.1: Dictionary from theory to measurement: the signal for each quantity, and the main pitfall in reading one as the other.

Quantity	Measured signal	Pitfall
density, occupation	in-situ absorption [41]; fluorescence [50, 64]	absorption gives the column density; the microscope sees the occupation modulo two [50]
momenta $n(\mathbf{k})$	TOF image [2]; focusing in 2D [119]	far field needs ≈100 ms; typical flights last 20 ms [2]
coherence $g_1(r)$	fringes [7, 124]; TOF peak width [99, 125]	peaks persist above T_c [19] and in the Mott insulator [99]
superfluid fraction f_s	c_x^2/c_y^2 [11]; continuity eq. [22]; Josephson [67]	a tensor; c_x^2/c_y^2 gives only the component along the modulation [11]
sound speed c_s	density perturbation [44]; Bragg slope [4, 12]	below T_c a 2D gas carries first and second sound [8]
$S(\mathbf{q},\omega)$	Bragg transfer [3, 62]	peak falls with depth faster than Bogoliubov [63]
gap, Higgs mode	modulation spectroscopy [24, 49]	theory: peaks also in incommensurate superfluids [127]
critical velocity v_c	heating behind a moving obstacle [5, 6]	vortices set $v_\mathrm{c} < c_\mathrm{s}$ [6]; lattices add dynamical instability [47]
compressibility κ	in-situ $n(\mu)$ with LDA [101]	superfluid and Bose glass are both compressible (QMC) [23]

Open

No Bragg measurement on a disordered or Bose-glass condensate was found in the literature checked for this map. Which signal of Table 3.1 separates a Bose glass from a weak superfluid is open: QMC uses winding numbers [23], experiments coherence [13].

Key points

- Quasi-2D needs $\hbar\omega_z \gg \mu, k_B T$; experiments reach $\tilde{g} \sim 0.1$ [32, 41].
- A TOF image at 20 ms is not the momentum distribution of a coherent cloud; test calculations on the image itself [2].
- Bragg spectra give $S(\mathbf{q}, \omega)$ and c_s; modulation spectra give gaps [4, 49].
- f_s comes from sound anisotropy, continuity or Josephson dynamics [11, 22, 67]; a calculation can reproduce the measured c_x, c_y and Leggett's formula.

4 Data, benchmarks and verification

A benchmark is a number that a calculation without free parameters and an experiment both give for the same system; the list is short, and every entry comes with conditions.

4.1 Numbers that theory and experiment share

We call a comparison a benchmark when the theory has no adjustable parameter and the experiment measures the quantity the theory computes, not a proxy for it. Table 4.1 collects the pairs for which both numbers were checked for this map (BKT, Berezinskii–Kosterlitz–Thouless; QMC, quantum Monte Carlo; DMRG, density-matrix renormalisation group; GPE, Gross–Pitaevskii equation).

Every row comes with conditions. The lattice experiments are trapped and at finite temperature, while the theory describes the ground state of an infinite lattice. In three dimensions the measured point lies 2σ from QMC [80]. Mark *et al.* give 34.8 and 29.3 only as theory values, "close to" each other, and measure number-dependent multi-body shifts [91]. In two dimensions the condensed fraction agrees with QMC "for a finite-sized 2D system" [130], and QMC of a trapped 2D gas with up to 576 000 bosons gives $T_{\mathrm{KT}} \simeq 0.70\, T^0_{\mathrm{BEC}}$ and a superfluid "with vanishing condensate fraction" [110]. The velocity row mixes geometries. Kwon *et al.* take 0.37, the limit $R \gg \xi$, from Gross–Pitaevskii work of 2000–2014 and, according to the paper, ascribe their "about 0.40" to finite R/ξ [16]; the 3D value 0.26 is Jackson *et al.*'s quotation of the experiment of Raman *et al.* [5, 76]. In the quasicrystal, QMC and experiment agree on the depth but through different observables: the winding-number f_s in the one, momentum-space coherence in the other [13, 23].

4.2 Pitfalls

A time-of-flight (TOF) image of a coherent lattice gas is not its momentum distribution. The far-field time $t_{\mathrm{FF}} \approx m\, l_c R_0/\hbar$, with l_c the coherence length and R_0 the cloud size, is about 100 ms for coherent samples by the authors' estimate, while typical expansions last 20 ms [2]. Sharp peaks also appear above T_c, where superfluid density and condensate fraction vanish (QMC, up to 1.4×10^4 particles) [19]. Pollet *et al.* hold that a proper TOF analysis "yields critical parameters accurately" while in-situ profiles are "barely suitable" near strongly correlated critical points; the peak-width criterion they discuss is one they cite ("It has been proposed …"), not their own [133]. Visibility is a weak signature too: it is "still around 30 %" at 15 E_{R} in a gas that is insulating beyond 12.5 E_{R} [99]. In one dimension the transition shows "most directly in the half width of the interference peak" [125].

A lattice critical velocity need not be Landau's. Burger *et al.* lost superfluidity at $v_{\max} =$

Table 4.1: Benchmarks with a checked number on both sides (in part where marked "as reported"). E_R: recoil energy of the lattice in question. Theory is homogeneous unless a trap is named; experiments are trapped or boxed, at finite temperature; v_c/c_s uses each paper's sound speed.

quantity	theory	experiment	source
Bose–Hubbard model, unit filling			
$(U/J)_c$, 3D cubic	QMC 29.34(2); mean field $5.8 \times 6 = 34.8$	34.2(2.0) at 13.5(2) E_R, "agrees with the mean field theory prediction"	[80, 88, 92]
$(U/J)_c$, 2D square	QMC $(J/U)_c = 0.05974(3)$; $U_c/J = 16.7424(5)$; mean field 23.31	condensed fraction vanishes (superfluid to normal) at $U/J = 15.8(20)$, $k_BT = k_B \times 33(4)$ nK	[90, 93, 94, 130]
$(U/J)_c$, 1D	DMRG $J_c/U = 0.277 \pm 0.01$, i.e. ≈ 3.61 (our inverse)	crossover to a discrete spectrum between $U/J = 4$ and 8 (tubes)	[49, 95]
Two-dimensional gas			
critical point	$D_c = \ln(C/\tilde{g})$, $C = 380 \pm 3$; μ_c constant $\xi_\mu = 13.2 \pm 0.4$	$D_c = 8.0(0.7)$ (written ρ_c) against $\ln(380/\tilde{g}) = 8.3$; trapped critical point matched "without any free parameters"	[60, 107, 108, 119]
superfluid jump	$n_s\lambda^2 = 4$	D_s from first and second sound jumps from 0 to 4; $T_c = 42(4)$ nK, $\tilde{g} = 0.64(3)$	[8, 32]
exponent at T_c	$\eta_{BKT} = 0.25$; finite-size formula about 0.21	$\eta_c = 0.17(3)$ in equilibrium; 0.13(1) after a quench	[42, 131]
sound, box, $\tilde{g} = 0.16(1)$	Bogoliubov 1.6(1), two-fluid 1.4(1), classical field 1.47 mm/s (as reported)	1.49(3) mm/s at $T/T_c = 0.37(12)$	[44, 132]
Excitations and flow			
Bragg, 3D trap	shift $4n_0g/7h$; Bogoliubov spectrum in the local density approximation	$(0.54 \pm 0.07)\, n_0g/h$; $\mu/h = 2.02(9)$ kHz (arXiv version), $c_{eff} = 2.0(1)$ mm/s	[3, 4]
v_c/c_s, moving obstacle	0.37 (2D, large cylinder); 0.45 ± 0.01 (2D, impenetrable cylinder); 0.13–0.20 (3D trap)	0.26 (3D); 0.11(2) and 0.07(2) (two densities); 0.3–0.6 (2D); about 0.40 (oblate)	[5, 6, 14–16, 76]
Superfluid fraction and localisation			
f_s tensor, triangular lattice	GPE; Leggett bounds of the measured profiles	$f_s^{(yy)} = 0.82(2)$ from the dynamic method at $V_0 \simeq 2.4\, g\rho_0$; the static method agrees; both agree with GPE and the bounds	[22]
f_s, 1D modulation	Leggett's bound is an identity; $f_{s,x} = c_x^2/c_y^2$	sound anisotropy against Leggett's formula on the measured density	[11]
Bose glass, 8-fold quasicrystal	QMC $V_c/E_R \simeq 1.76 \pm 0.01$; $f_s = 0$ in the glass "except close to the separation line"	non-interacting localisation at 1.78(2) E_R	[13, 23, 56]
3D mobility edge	self-consistent theory; numerics	mobility edge $E_c = 52(5)$ nK at speckle strength $V_R = 46.8(1.4)$ nK; about 20 % (40 %) above	[52]

5.3(5) mm/s against a local $c_{s,\max} \approx 5.2$ mm/s and wrote that it "breaks down first in the low-density regions", reading the loss as "the spontaneous emission of elementary excitations" without naming the Landau criterion [45]; Wu and Niu showed that a lattice condensate can also be dynamically unstable, and suggested this instability for the dissipation Burger *et al.* saw [17]. Fallani *et al.* attributed lifetimes falling from about 10 s to 10 ms at $q \approx 0.55\, q_B$ "unambiguously" to dynamical instability [47]; De Sarlo *et al.* found an almost pure condensate (< 20 % thermal) insensitive to the lattice velocity over 15 s, whereas a cloud with $\simeq 35$ % thermal fraction decayed even at small velocity, the energetic instability acting in (416 ± 55) ms [18].

Lattice spectra leave Bogoliubov theory behind. In a 3D lattice the Bragg resonance falls with

depth faster than Bogoliubov theory predicts; the two differ most at 3.5 E_R, and the authors tentatively blame quantum depletion [63]. 1D superfluids show a continuum that "calls for an interpretation beyond the Bogoliubov spectrum" [49]. Time-dependent DMRG finds sharp modulation peaks in incommensurate superfluids as well, and it matches the measured positions "provided we assume the presence of an incommensurately filled region" [127]. The GPE-based truncated Wigner approximation (TWA) "is not suitable to formulate those fluctuations in the strongly interacting limit" [134]; its SU(3) successor deviates noticeably from a 2D quench experiment even at short times, according to a preprint [135].

Key points

- Measured $(U/J)_c$: 34.2(2.0) in 3D (mean field; 2σ from QMC); in 2D the condensed fraction vanishes at $U/J = 15.8(20)$, a superfluid-to-normal transition at $k_B \times 33(4)$ nK.
- BKT: $C = 380 \pm 3$ and $n_s\lambda^2 = 4$ hold without free parameters; $\eta_c = 0.17(3)$ lies below 0.25 and below the finite-size estimate of about 0.21.
- Measured v_c/c_s spans 0.07–0.6; lattice thresholds can be instabilities.
- The measured f_s of Chauveau *et al.* and Rabec *et al.* come from weakly interacting condensates and agree with Leggett's bound, Rabec's also with GPE [11, 22].

5 State of the art

Apart from stirring and transport, each line measured coherence first and superfluid response later; the measurements of the superfluid fraction in modulated gases found for this map are recent and come from three groups.

The experiments fall into three lines: two-dimensional gases, lattice (and three-dimensional) superfluids, and disordered or quasi-periodic potentials. Figure 5.1 sorts them by observable, Fig. 5.2 places them in time and Table 5.1 lists the groups; E_R is the recoil energy of the lattice in question.

5.1 Two-dimensional gases

The two-dimensional (2D) line began with coherence. Interfering two planes, Hadzibabic *et al.* saw the correlation exponent drop from about 0.5 to 0.25 on cooling, and "the loss of long-range coherence coincides with the onset of proliferation of free vortices" [7]. Cladé *et al.* produced a gas that "is quasi-condensed but is not superfluid" [20]; Tung *et al.* saw the compressibility grow to 1.7 times their reference value κ_{ba} before a momentum spike at $p = 0$ appeared [119]; thermodynamics showed scale invariance [41, 59]. Theory for the trap kept pace. Semiclassical profiles match QMC above T_{KT} [106], the critical phase-space density is ∼8 and "close to 10" in two experiments [111], and classical fields show coherence emerging above T_{BKT} with the exponent 0.25 near the trap centre at T_{BKT} [112]. Sunami *et al.* measured the exponent at the transition [42], then quenched by splitting the gas: phonons thermalise fast, vortices unbind slowly with $n_v \propto t^{2\nu}$, $\nu \sim 1$, through "a metastable supercritical phase" [131].

Response measurements came with uniform traps, first a 3D box whose $T_c = 92 \pm 3$ nK agrees "with the prediction for a uniform Bose gas" [136], then 2D boxes with Kibble–Zurek defect scaling [43]; homogeneity "almost invariably makes the interpretation of the experiments and the comparisons with theory more direct" [137]. A 2 μm obstacle stirred at $\tilde{g} = 0.093$ gave $v_c/c_s = 0.3$–0.6, with frictionless flow above $\mu_{loc}/k_BT \approx 0.24$ [6]; classical fields trace the

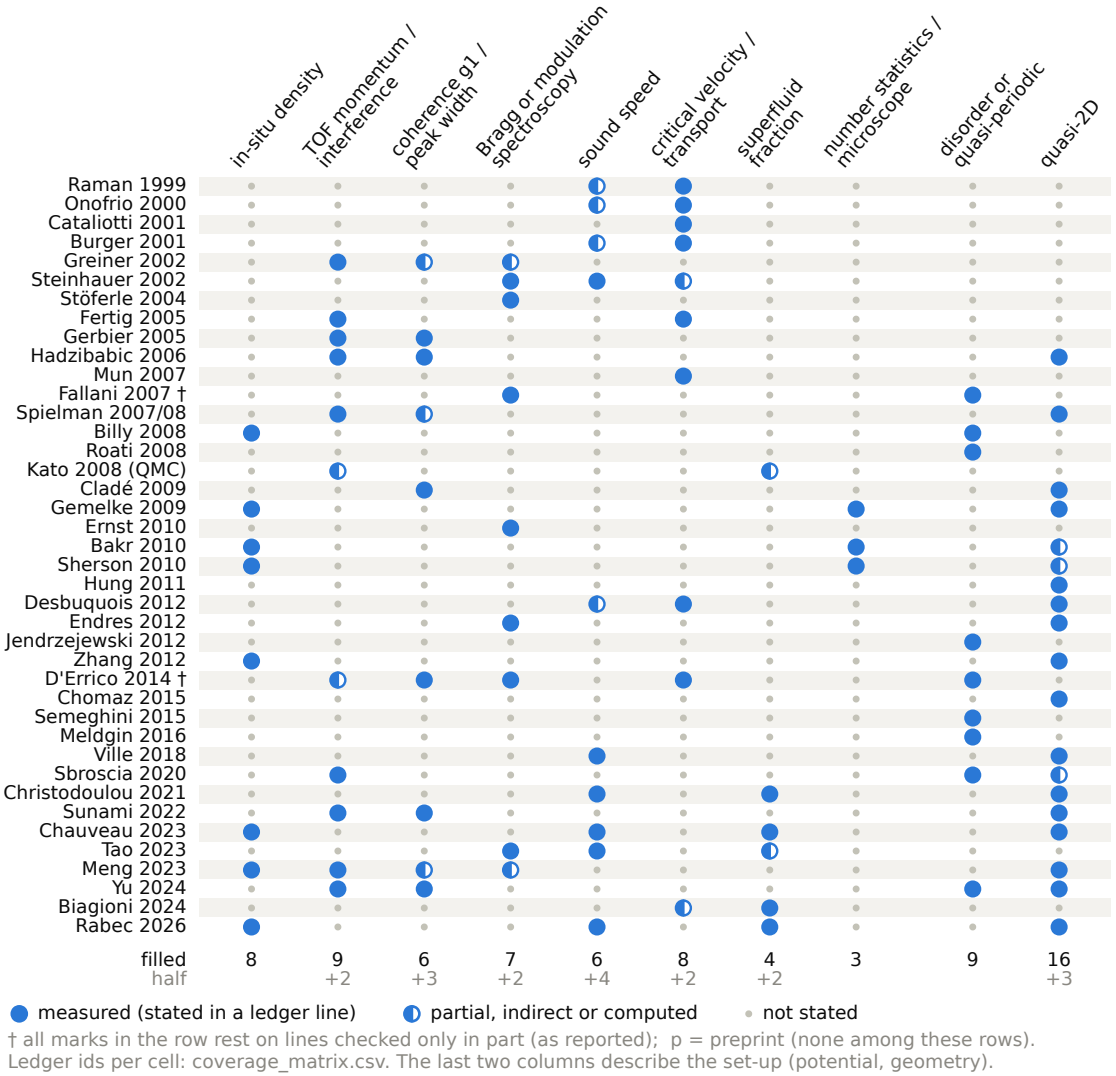


Figure 5.1: Coverage matrix: experiments of this map (rows) against observables (columns). Full dot: measured, as stated in a checked ledger line; half dot: partial, indirect or computed; small dot: not stated in the lines read, which does not mean never measured. Rows marked † rest on lines checked only in part. The row Kato 2008 (QMC) is a calculation, shown for its computed interference peaks.

heating onset to vortex–antivortex pairs and reproduce the data "if the wing temperature is used, rather than T_{eq}" [77]. Theory had predicted for $\tilde{g} = 0.1$ (their g) sound jumps $c_1/c_0 = 2.36 \to 1.94$ and $c_2/c_0 = 0.56 \to 0$ at the transition, with second sound "easily excited through a density perturbation" [138], and later calculations, not checked for this map, treat collisionless sound [139] and stronger interactions [140]. In a box the sound speed was measured, with strongly damped density waves above T_c [44], and $c(T)$ below T_c follows "a Bogoliubov estimate that includes the superfluid density at nonzero temperature" [132]. In ^{39}K at $\tilde{g} = 0.64(3)$ first and second sound were resolved below T_c; above it second sound "is replaced by a diffusive, overdamped mode", and the superfluid density follows BKT theory with its universal jump (speeds in figures only) [8]. The classical field of a 2D gas carries an SO(2,1) symmetry, with breathers seen at $T/T_c < 0.3$ [141]. f_s of a modulated 2D gas has been measured since 2023, under a 1D modulation of period 3.93(4) μm [11] and in a triangular lattice of period 6.0(1) μm, where the tensor is "nearly isotropic" [22].

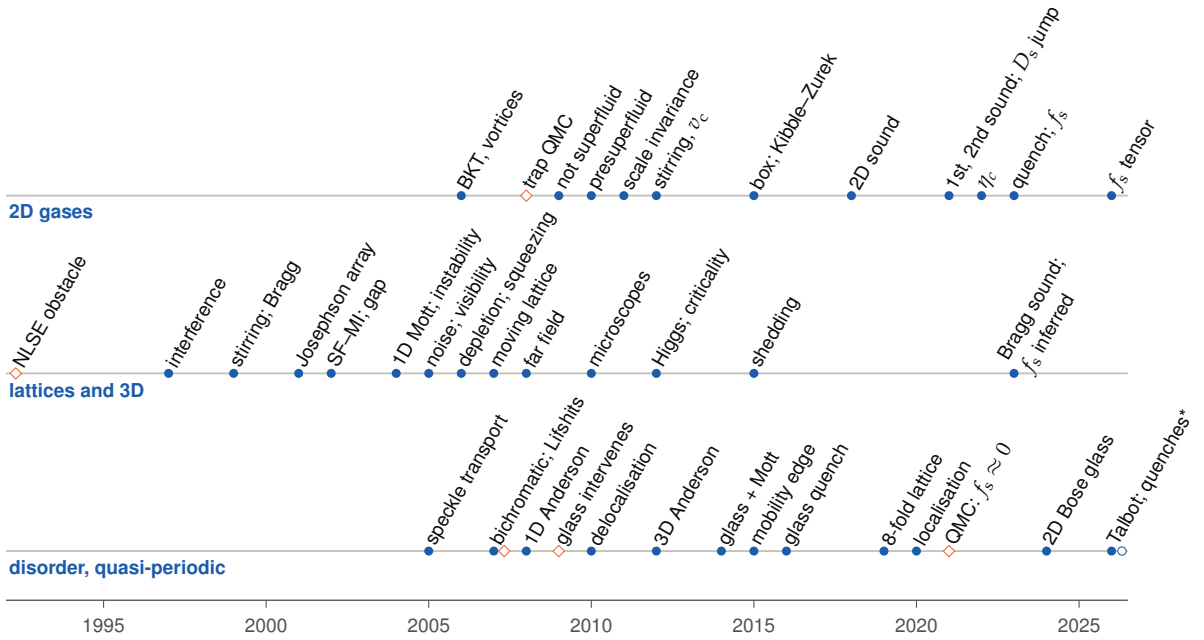


Figure 5.2: Timeline, one mark per year and line. Dots: experiments; open diamonds: theory (the nonlinear-Schrödinger obstacle problem, 1992; trap QMC, 2008; Lifshits glass, 2007; Bose glass always between insulator and superfluid, 2009; QMC $f_s \approx 0$, 2021); open dot and asterisk: preprint. 2D: [6–8, 11, 20, 22, 42–44, 59, 110, 119, 131]. Lattices and 3D: [2, 3, 5, 12, 16, 24, 29, 46, 47, 49, 50, 64, 75, 80, 99, 102, 103, 118, 121, 122]. Disorder: [9, 10, 13, 21, 23, 51–53, 55–57, 65, 66, 142–147].

5.2 Lattice and three-dimensional superfluids

The superfluid–Mott-insulator (SF–MI) transition is still characterised mostly through coherence and gaps. In the 3D lattice "at a potential depth of around 13 E_r the interference maxima no longer increase in strength", with $U/J \approx 36$ there; a potential gradient makes tunnelling resonant when neighbouring sites differ by U, two narrow resonances stand at 20 E_R "on top of an otherwise completely flat excitation probability", the second at exactly twice the first, and coherence is "fully visible again" 4 ms into a ramp-down, with steady peaks after 14 ms [29]. Mean-field theory puts the $n = 1$ tip at $U/zJ \approx 5.83$, the spectrum going "from a gapped to a gapless phase" [89, 90]. A modulated 3D superfluid showed "an unexpected resonance" above the first Bogoliubov band [148]; depletion beyond 50 % appears as a diffuse TOF background, semi-quantitatively Bogoliubov [103]; suppressed spin-changing collisions give "direct evidence for the emergence of number-squeezed states", in the insulator up to $6.0(3) \times 10^4$ atoms, where a doubly occupied plateau should form [102]. In situ, experiments measured the suppressed compressibility of Mott domains [101] and quantum-critical scaling, with a critical chemical potential of $-4.5(6)J$ against $-4J$, dynamical exponent $z = 2.0(3)$ and $\nu = 0.54(5)$ [118]. Above 18 E_R the U and $2U$ resonances shift down through higher-band admixture [91].

Transport probes the superfluid itself rather than its coherence. About 200 coupled condensates oscillate in phase, undamped, and lose coherence through a dynamical instability at large displacement [46]; a ring gives $v_c/c_{\text{eff}} \approx 0.6$ for all barriers [38]. In oblate condensates an obstacle nucleates vortex dipoles at $v_c \sim 0.1c$ [36] and sheds vortices at v_c/c_s about 0.40, with a dip at $V_0 \approx \mu$ [16], in a von Kármán street above $v_c = 1.11(5)$ mm/s at $c_s \approx 5.1$ mm/s [37].

The excitations are mapped by spectroscopy. Lattice modulation gives a continuum in the superfluid and discrete Mott resonances, the second at (1.91 ± 0.04) times the first in 1D tubes, where Greiner's 3D tilt gave exactly twice [49]. Theory predicts two gapped modes in the insulator and, in the superfluid, sound plus a gapped (Higgs) mode by which the "puzzling

finite frequency absorption observed in the superfluid phase could be explained" [149], and a Bragg response that deep in the Mott regime "is temperature dependent" [150]. Bragg spectroscopy resolved up to three bands, closer to Bogoliubov than to Bloch bands below 6 E_R [48], momentum-resolved spectra at 0.64, 0.71 and 0.93 π/a [62], and the loss of response in the insulator [61]. The amplitude mode appeared at about 650 Hz above a 200 Hz sound peak [128], its onset softening towards $j_c \simeq 0.06$ [24]. QMC at $(U/J)_c = 16.7424$ finds a Higgs resonance that is "no longer seen as a resonance peak for a detuning as small as 20 %" and explains why the onset protocol works [151]; the longitudinal susceptibility "can completely obscure the amplitude peak", the scalar one exposes it [152].

Table 5.1: Groups whose experiments this map cites, named by senior authors on the papers' bylines (the last row is theory); places are omitted because affiliations were not checked. Lines: 2D gases (2D), lattices and 3D gases (L), disorder and quasi-periodic (D), superfluid fraction (f_s).

group (senior authors)	lines	papers cited in this map
Dalibard, Beugnon	2D, f_s	[6, 7, 11, 22, 41, 43, 44, 113, 141]
Hadzibabic	2D, L	[8, 60, 136]
Chin	2D, L	[59, 101, 117, 118]
Cornell; Foot	2D	[119]; [42, 114, 131]
Phillips, Porto, Spielman, Campbell	L, 2D, f_s	[12, 20, 35, 38, 79, 123, 130]
Ketterle	L	[3, 5, 14, 39, 40, 80, 103, 122]
Shin; Anderson	L	[16, 37]; [36]
Davidson (Steinhauer)	L	[4, 58]
Bloch	L, D	[2, 24, 29, 64, 99, 100, 102, 121, 153–155]
Esslinger	L	[49, 120, 124, 148]
Inguscio, Fort, Modugno	L, D, f_s	[10, 18, 45–48, 52, 53, 61, 65–67, 142, 156, 157]
Aspect, Bouyer	D	[9, 51, 143]
DeMarco; Schneider; Kuhr	L, D	[54, 147, 158]; [13, 55–57]; [21]
Sengstock; Greiner; Nägerl; Heinzen; Schneble	L, D	[62, 128]; [50]; [91]; [63]; [159]
Mathey (theory)	2D	[77, 132]

5.3 Disorder and quasi-periodic potentials

The first speckle experiments, in 2005, stopped transport without Anderson localisation. Strong disorder trapped the condensate in deep wells, weaker disorder gave stripes after expansion and strongly damped dipole and quadrupole modes [142]; expansion stopped at high disorder, which the papers ascribe (not checked for this map) to a healing length below the grain spacing [143] and to deep wells acting as single traps [156]. A review calls this "not Anderson localization, but a classical localization that can be explained with simple energetic arguments" [160]. Theory set the conditions. An expanding condensate localises exponentially when its initial healing length exceeds the speckle correlation length and algebraically otherwise, as the paper states [161]. According to its abstract, weak interactions leave the gas in "a finite number of localized single-particle Lifshits states", while strong ones form a delocalised disordered condensate [144]. Bogoliubov quasiparticles localise too, screened in the phonon regime, and the authors expect this to be "observed directly, for instance as a broadening of the resonance lines in Bragg spectroscopy" [162]. In 2008 localisation was seen in 1D speckle beyond "an effective mobility edge", algebraic otherwise, with a theory "with no adjustable parameters" getting the localisation length right in magnitude and shape [9]; in a bichromatic lattice, with Δ the strength of the second lattice, the Aubry–André threshold $\Delta/J = 2$ is broadened and shifted, with onset at $\Delta/J \cong 6$ and exponential localisation for $\Delta/J \geq 7$ [10]. The 3D mobility edge was measured

"in the disorder–energy plane" [51, 52]; reviews cover the period [30, 129, 163].

With interactions the question becomes the Bose glass, "an insulating but compressible state without long-range phase coherence" [13, 28]. With bounded disorder "the SF-I transition is always to the compressible Bose glass phase" [146]; in 2D at unit filling the glass separates superfluid and insulator, which meet only at the clean $U_c/J = 16.7424(5)$, and superfluidity survives to a disorder bound $\Delta/J = 70$ [94]; in 1D a finite-temperature transition separates fluid and insulator [164]. In 2D speckle $mc^2 = f_s(n\kappa)^{-1}$, κ in the authors' normalisation, ties sound to f_s, the Leggett bounds being "too loose" in 2D but coinciding in 1D [165]. In the laboratory, repulsion delocalises a quasi-periodic gas, the crossover centred at an interaction energy $E_{\mathrm{int}} = 0.05\Delta$ [53], with subdiffusion $\sigma \propto t^{\alpha}$, $\alpha = 0.2$–0.4 [157]; at strong interaction "a strongly-correlated Bose glass coexisting with a Mott insulator" appears [66]. Transport at unit filling excludes "a disorder-induced or 're-entrant' superfluid" [54]; quenches found evidence for "superfluid puddles in the Bose glass phase" [147]; many-body localisation was explored in 2D [155].

Quasicrystals gave the 2D Bose glass. An eightfold lattice produced self-similar diffraction [55] and non-interacting localisation at 1.78(2) E_R [56]. QMC finds $f_s = 0$ "except close to the separation line" [23]; in 1D, where $V_c/E_R \simeq 1.38 \pm 0.01$, "the SF fraction is strictly zero" at sizable compressibility, as reported [166]; at a 2D coupling $\tilde{g}_0 = 0.86$ the glass survives to $T \simeq 0.7\,E_R/k_B$ [167], and trapped systems have been simulated [168]. Experiments found that "it is not possible to adiabatically traverse the Bose glass on typical experimental timescales" [13], with quench timescales set by quasi-disorder or tunnelling (preprint) [57]; microscopy and Talbot interferometry of about 200 atoms in a lattice with programmed box disorder show the coherence length falling from 15(3) to 1.4(3) sites and an Edwards–Anderson parameter $q \approx 0.06$ of a possible 0.25 [21].

Key points

- 2D: sound gave the jump of D_s from 0 to 4 (2021); f_s under a modulation from 2023.
- Lattices: Mott resonances at U and $2U$ in 3D, the second at 1.91(4) times the first in 1D; the QMC Higgs peak is gone at 20 % detuning.
- Disorder: classical localisation in 2005, Anderson localisation in 2008; QMC gives $f_s \approx 0$ in the Bose glass except close to the separation line.

6 Position: interference certifies coherence, not superfluidity

A sharp time-of-flight peak shows that phases are correlated over some distance; only a response to flow, a twist, a sound wave or a modulation shows that the gas is superfluid.

The claim covers weakly interacting bosons in periodic, modulated or disordered potentials and in two dimensions (2D). It concerns what an observable certifies, not what the phases are. Figure 6.1 sets it out in three columns and a decision box; the last column applies it to the twisted-bilayer optical lattice (TBOL; Table 6.1).

What interference measures. Fringes between two independent condensates, first reported by Andrews *et al.* [122] (seen in listings only), record a relative phase. Bloch, Hänsch and Esslinger measured the spatial correlation function with a radio-frequency double slit: above T_c "a rapid gaussian decay, as expected for a thermal gas", below T_c "a slow decay towards a plateau" [124]. For condensates without true long-range order, Polkovnikov, Altman and Demler showed

that the interference pattern "contains information about correlation functions within each condensate" [126]. Hadzibabic *et al.* used this in a 2D gas: the exponent α of the contrast decay is close to 0.5 at high temperature and "rapidly drops to about 0.25" on cooling, and "the loss of long-range coherence coincides with the onset of proliferation of free vortices" [7]. The superfluid density enters only through Berezinskii–Kosterlitz–Thouless (BKT) theory, in which the universal jump "corresponds to a sudden drop in α from 0.5 to 0.25" [7]. In a lattice the superfluid gives sharp peaks, while in the Mott insulator (MI) "exact numbers of atoms are localized at individual lattice sites, with no phase coherence across the lattice" [29]. In a disordered 2D lattice, Talbot interferometry gives a coherence length that falls from 15(3) to 1.4(3) lattice sites at the strongest disorder [21].

Where it misleads. Five results separate coherence from superfluidity. Quantum Monte Carlo (QMC) of the three-dimensional (3D) Bose–Hubbard model finds "that sharp peaks are not a reliable diagnostic of superfluidity": above T_c, "where both superfluid density and condensate fraction vanish, the interference pattern can nevertheless have sharp peaks riding over a broad background" [19]. A quasi-2D gas can be bimodal, with a coherence length longer than the thermal cloud's, and still be "quasi-condensed but is not superfluid" [20]. A time-of-flight (TOF) image is the momentum distribution only in the far field, and for phase-coherent samples "the far-field condition is usually not met": $t_{\mathrm{FF}} \approx 100\,\mathrm{ms}$ against the usual 20 ms for a lattice spacing near 400 nm [2]. In the MI "phase coherence on short length scales persists", from "a coherent admixture of particle-hole pairs"; the contrast is "still around 30%" at 15 E_{R} and drops to a few percent only at 30 E_{R}, decaying as a power of U/zJ with an average exponent $\alpha = -0.98(7)$, as reported [99, 100]. The Bose-glass experiment in an optical quasicrystal states its criterion plainly: "A high-contrast interference pattern composed of sharp peaks indicates the presence of long-range phase coherence, the signature of the superfluid phase" [13]. This is a coherence criterion; it works there because QMC finds $f_{\mathrm{s}} \approx 0$ in the glass, except close to the separation line [23]. Pollet, Prokof'ev and Svistunov argue the other side: a proper analysis of TOF images "yields critical parameters accurately", whereas in-situ profiles are "barely suitable" for critical points [133]. TOF locates a transition when compared with a calculation of the same image.

What certifies superfluidity. Seven responses are in use, each read through its own model. (i) *Critical velocity.* Stirring gives a threshold below which the gas does not heat, but always below Landau's $v_{\mathrm{c}} = c_{\mathrm{s}}$: about 1.6 mm/s against a peak sound speed of 6.2 mm/s in 3D [5], $(0.11 \pm 0.02)\,c_{\mathrm{s}}$ at the higher of two densities [14], $\sim 0.1\,c$ for a single vortex dipole in an oblate condensate [36], "about 0.4" for the stiffest obstacle in another one [16], and 0.3–0.6 in a 2D gas [6]. Vortex nucleation sets these values: Gross–Pitaevskii (GP) theory gives $(0.45 \pm 0.01)\,c$ for an impenetrable cylinder in 2D [15], $0.37\,c_{\mathrm{s}}$ for a large cylinder in the works cited by Kwon *et al.* [16], and 0.13–0.20 c_{s} in a trapped 3D condensate, where "vortex lines first appear" at the low-density edge [76]. In a lattice, flow can also fail by a dynamical instability, whose region lies inside the Landau region [17]; both were characterised in one experiment, where a thermal fraction "triggers" the energetic one [18]. Earlier, Burger *et al.* had read the breakdown as the emission of elementary excitations at the local sound speed, naming neither the Landau criterion nor an instability [45]. Wu and Niu suggested a dynamical instability [17]; later moving-lattice experiments identified dynamical instability [47] and two-atom scattering [80] in their own data. (ii) *Moving frame and phase twist.* Leggett bounded f_{s} by the density modulation [25] and treated it for an arbitrary many-body system at $T = 0$ [169], as summarised in [11, 26]. In the words of Pérez-Cruz, Astrakharchik and Massignan, "The superfluid fraction f of a quantum fluid is defined in terms of the response of the system to a weak and constant drag", with $E(v) = E(0) + f_{\mathrm{s}} N m v^2/2$ as $v \to 0$ and 2D brackets $f^- \le f_{\mathrm{s}} \le f^+$ from the density alone [26]. For 2D speckle the brackets are "too loose of an estimate", while "in the 1D case, the upper and lower Leggett bound coincide" [165]. (iii)

Sound. Across a one-dimensional (1D) modulation $f_{s,x} = c_x^2/c_y^2$, and "the bound reduces to an identity" for a weakly interacting condensate [11]. In a triangular lattice $f_s^{(yy)} = 0.82(2)$ at $V_0 \simeq 2.4\, g\rho_0$ from the compressibility and the sound speed through $c^2 = f_s/(m\kappa)$, consistent with the static method, GP simulations and the Leggett bounds of the measured profiles [22]; an analogous relation (κ per particle), $mc^2 = f_s(n\kappa)^{-1}$, is proposed for disordered gases [165]. In a 1D lattice "the speed of sound saturates Leggett's bound", a statement resting on Bogoliubov–de Gennes (BdG) theory [12]. (iv) *Second sound.* In a 2D box of ^{39}K ($\tilde{g} = 0.64(3)$, $T_c = 42(4)$ nK) the second sound becomes "a diffusive, overdamped mode" above T_c, and the superfluid phase-space density deduced from both sound speeds and the 2D equation of state jumps "from 0 to 4 at the critical point" [8]. (v) *Josephson oscillations.* Single supersolid cells gave "a relatively large sub-unity superfluid fraction" [67]; in a lattice array, small oscillations are coherent and undamped, and large displacements make the array dynamically unstable [46]. (vi) *Winding numbers.* QMC samples the twist response through windings [83]. In an eightfold quasicrystal "we find $f_s = 0$, except close to the separation line" in the Bose glass, while the compressibility crosses over smoothly [23]; in the disordered 2D Bose–Hubbard model the glass "always separates SF and MI", which meet only at $U_c/J = 16.7424(5)$ [94]. (vii) *Amplitude (Higgs) mode.* Modulating a 2D lattice depth by 3% gives an onset frequency that softens towards the critical ratio $j_c = (J/U)_c \simeq 0.06$ [24]; QMC finds a Higgs resonance near the critical point that "is no longer seen as a resonance peak for a detuning as small as 20%" [151]. This response tests the order parameter's amplitude near the MI, not the stiffness.

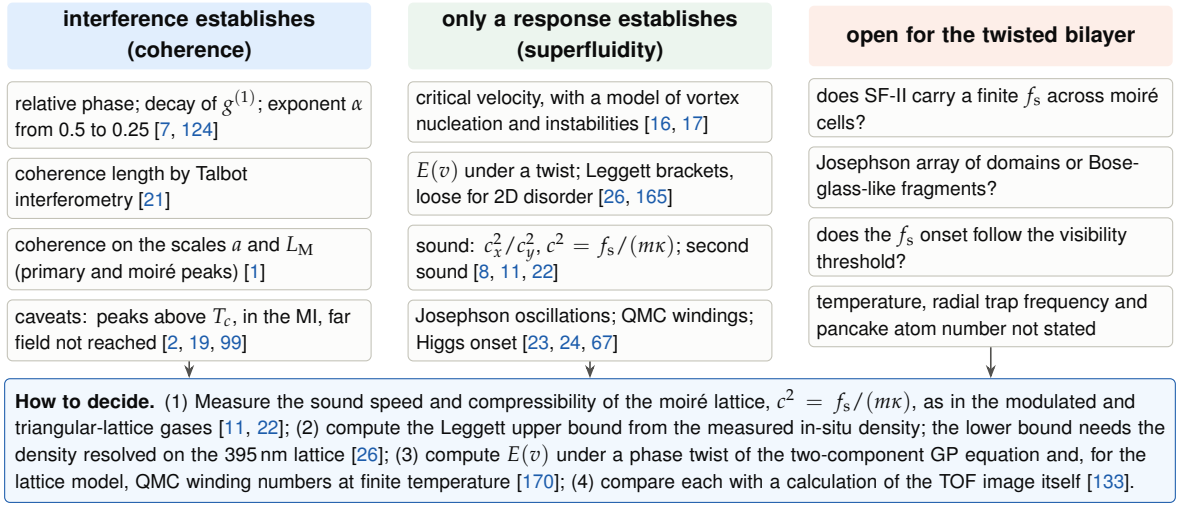


Figure 6.1: Position diagram. Left: what interference establishes; middle: responses that certify superfluidity, each resting on a model (vortex nucleation, BdG, GP, BKT theory, QMC); right: what is open for the TBOL phase diagram (nothing there has been measured); bottom: what would decide.

The case. The one TBOL experiment with bosons found for this map builds its phase diagram from the visibilities of two families of TOF peaks, at the primary and at the moiré momenta [1]. SF shows both; SF-II, "superfluid only with short-range coherence", keeps the primary peaks only, and the Extended Data describe "superfluid domains embedded in a gapped insulator". Along path III, which raises the microwave (Rabi) coupling Ω_R between the layers at a fixed depth of 8 or 14 E_R, "the visibility at the moiré momenta presents the threshold behavior and emerges at $\Omega_R > 0.5E_r$" (the paper writes E_r for E_R). In SF, phases are correlated over more than one moiré period, $L_M = 4.35\,\mu$m; in SF-II, over the primary lattice but not across cells. It is open whether SF-II carries a finite f_s across the sample, as a Josephson array of domains would, whether it is closer to a glass with $f_s \approx 0$, as in the quasicrystal [23], or whether temperature has depleted thin superfluid domains, as QMC finds for a single-component twisted potential [170].

Because no temperature was found in the paper, we cannot exclude the third reading.

Table 6.1: What the visibility-based TBOL diagram says [1] and what a response measurement would say (outcomes that would be informative; none has been measured).

Phase	Visibility-based diagram	Response measurement
SF	both peak families: coherence beyond one moiré period	the f_s tensor from sound and compressibility or a twist (triangular lattice: 0.82(2) [22])
SF-II	primary peaks only: coherence within moiré cells, not across them	finite f_s with a soft inter-domain Josephson mode (array), or $f_s \approx 0$ with finite compressibility (glass-like) [23]
MI	no peaks	suppressed compressibility in situ [101]; a gap [29] and a discrete modulation spectrum [49]; $f_s = 0$
insulator I	no peaks; filling modulated at the moiré scale	gapped or compressible: separates an insulator from a Bose glass [13, 94]
boundaries	threshold of a peak visibility, e.g. $\Omega_R > 0.5\, E_R$ on path III	onset of f_s along each axis, which at $T > 0$ need not coincide with the peak threshold [19]

Limits of the thesis. The thesis concerns a visibility read alone. TOF analysed against a full calculation of the image locates critical points [133], the first "ab initio comparison between experiments and quantum Monte Carlo" exists for lattice bosons [153], and Talbot coherence lengths read with site-resolved images (Edwards–Anderson parameter $q \approx 0.06$) which the authors attribute to the Bose-glass phase [21]. Where theory ties coherence to superfluid density, the two coincide. In a uniform 2D gas the loss of coherence tracks vortex proliferation [7], the exponent measured at the transition, $\eta_c = 0.17(3)$, lies below 0.25 and below the finite-size estimate of about 0.21 [42], and sound confirms the universal jump [8]. Finally, in a sample of a few moiré cells coherence and f_s are both finite-size quantities, so neither separates a phase from a crossover without a size dependence.

Key points

- TOF peaks and visibilities persist above T_c and in the MI; far field is often not reached [2, 19, 99].
- Measured v_c lie between about 0.07 and 0.6 c_s because vortices nucleate first [16, 76].
- f_s was read from sound (0.82(2), triangular lattice) and Josephson oscillations; Leggett brackets are loose for 2D disorder [22, 67, 165].
- TBOL's SF and SF-II are defined by peak visibilities; whether SF-II carries a finite f_s across moiré cells is untested.

7 Around the topic

Supersolids, coupled condensates, flat-band lattices and photonic lattices ask the same question as a modulated condensate, which part of the fluid is superfluid, with one ingredient pushed to a limit.

7.1 Supersolids and Josephson physics

A supersolid is a condensate modulated by its own interactions instead of an external potential. The flow still has to pass through the density minima, so Leggett's upper bound [25], computed from the density profile alone (as summarised in [11, 26]), applies unchanged (our remark). Biagioni *et al.* measured f_s of a supersolid from the Josephson oscillations of single cells and

found "a relatively large sub-unity superfluid fraction" [67], a third route to f_s beside sound speeds and static density profiles (Chapter 3).

Blakie treated the two-dimensional supersolid in the Gross–Pitaevskii picture, in which "The superfluid fraction is described by a rank-2 tensor" and nonclassical translational inertia is tied to the effective mass [81]; Rabec *et al.* measured the same kind of tensor in an external triangular lattice (our remark). Beyond mean field, for stripes of dipolar bosons, quantum Monte Carlo (QMC) with diffusion and path-integral ground-state Monte Carlo find "The superfluid fraction along the stripes direction is always close to 1, while in the Y direction decreases to a small value that is nevertheless different from zero", and a total near $1/2$ close to the critical point [171]; the paper makes no statement about Leggett's bound. A preprint by Pizzino *et al.* is listed as finding Leggett's bound reliable for strongly interacting bosons at the crossover from two to one dimensions (not checked for this map) [86].

A deep lattice is an array of weak links. Cataliotti *et al.* made this literal with about 200 wells of about 1000 atoms each, in which "The square of the small-amplitude oscillation frequency is proportional to the microscopic tunneling rate"; at large displacement the array "becomes dynamically unstable, and the phase coherence is lost" [46]. The target system of Chapter 8 adds a coupling between internal states. For two coherently coupled condensates Abad and Recati showed that a finite coupling changes the first-order demixing transition into a second-order one [172]. Zenesini *et al.* observed "bubble nucleation in isolated and coherently coupled atomic superfluids", a false-vacuum decay [173]. Tunnel-coupled one-dimensional gases are Josephson systems too; in thermal equilibrium they act as "a quantum simulator of the sine-Gordon model" [174]; with Raman coupling they would show a commensurate–incommensurate transition "described by the Pokrovsky-Talapov model for the relative phase", with "a soliton staircase in the phase difference" [175]. Relaxation in such a junction was modelled by "stochastic classical field simulations using Gross-Pitaevskii equations in three spatial dimensions", which "reproduce the main experimental findings"; the relaxation persists "even at zero temperature" [176].

7.2 Flat-band bosons and photonic analogues

For a weakly interacting condensate in a periodic potential, Chauveau *et al.* state that the superfluid fraction "exactly coincides with the ratio m/m_x^*" [11], m^* being the band mass that also sets the sound speed [85]. A flat band is the limit $m/m^* \to 0$, so any superfluidity there must come from interactions and the geometry of the Bloch states. Julku, Bruun and Törmä showed for the kagome lattice that the sound velocity $c_s = (2Un_0/3)\sqrt{g(k_c)}$ is linear in U and that the depletion stays finite as $U \to 0$, set by the quantum geometry alone [177]. A 2026 preprint sharpens the conditions: "There can therefore be no superfluidity in a single flat band at a TRIM where the Bloch state has uniform densities" (TRIM: time-reversal-invariant momentum), and a stable two-dimensional flat-band superfluid "requires at least three bands within Bogoliubov theory" [178].

Flat-band experiments with bosons exist, but none of those read for this map measures f_s. In a kagome lattice, ^{87}Rb condensates show an interaction-enhanced group velocity in the flat band that agrees "quantitatively with predictions of the lattice Gross-Pitaevskii equation" [179]. Paraxial light in a nonlinear medium obeys a cubic Schrödinger equation, so a photonic lattice is a mean-field condensate without temperature or superfluid fraction. Photonic moiré lattices were used to study the localisation and delocalisation of light (from the title of [180]), and the formation of solitons controlled by the twist angle has "threshold properties that are a pristine direct manifestation of flat-band physics" [181]. These experiments test the single-particle and nonlinear-wave parts of the twisted-lattice problem (Chapter 8), not its superfluid response.

Key points

- Supersolid: "relatively large sub-unity" f_s from Josephson oscillations; QMC for dipolar stripes: f_s close to 1 along, small but nonzero across.
- Josephson array: squared small-amplitude frequency ∝ tunnelling rate; coherent coupling turns demixing from first to second order.
- Flat bands: a stable two-dimensional superfluid "requires at least three bands" within Bogoliubov theory (preprint); no flat-band or photonic experiment found measures f_s.

8 Observables for the twisted-bilayer problem

Each observable of the earlier chapters set against the one twisted-bilayer experiment with bosons found for this map: what it measured, what it left out, and what each reference measurement would add.

Table 8.1 gives, per observable, whether the twisted-bilayer optical lattice (TBOL) experiment measured it, what it certifies (Chapter 6), a template experiment and the calculation to compare.

8.1 What the experiment measured and what it left out

Meng *et al.* load ^{87}Rb into two square lattices of constant $a = 395\,\mathrm{nm}$, each felt by one hyperfine state, $|1\rangle = |F{=}1, m_F{=}1\rangle$ or $|2\rangle = |F{=}2, m_F{=}0\rangle$, twisted by $\theta = 5.21°$; the moiré period is $L_\mathrm{M} = 4.35\,\mu\mathrm{m}$, eleven lattice constants [1]. A 6.8 GHz microwave couples the layers with a Rabi frequency Ω_R up to about 1 E_R, with $E_\mathrm{R} = h \times 3.67\,\mathrm{kHz}$ (written E_r in the paper). An accordion lattice holds a pancake with "$\omega_z \approx 2\pi \times 1\,\mathrm{kHz}$". The data are microwave spectra at $V_0 = 0, 4$ and $10\,E_\mathrm{R}$, in-situ images at 50× magnification that resolve the moiré density, and state-selective momentum distributions after 18 ms of time of flight (TOF). At $4\,E_\mathrm{R}$ the calibrated coupling is $2\pi \times 893\,\mathrm{Hz} = 0.243\,E_\mathrm{R}$, or $\Omega_R/J = 2.07$, and the paper quotes $U/J = 1.67$ (its tie to that depth was not seen in the checks). Deepening the lattice at $\Omega_R = 0.24\,E_\mathrm{R}$ (path I) takes the gas from the superfluid (SF) through SF-II to the Mott insulator (MI); raising Ω_R at 8 or $14\,E_\mathrm{R}$ (path III) reverses the sequence, with the moiré visibility emerging at $\Omega_R > 0.5\,E_\mathrm{R}$. The paper's theory is a two-component Gross–Pitaevskii equation (GPE) with "$g = g_{11} = g_{22} = g_{12}$" and a single-site Gutzwiller ansatz, a product of on-site states expanded in the Fock basis.

Three inputs that every observable below depends on were not found in the paper: the temperature, the radial trap frequency and the atom number in the pancake (the condensate holds up to 5×10^5 atoms). A fourth, the scattering lengths of the pair $|1,1\rangle, |2,0\rangle$, was not looked up for this map; measured values exist for another pair, $a_{12} = 98.006(16)\,a_0$ and $a_{22} = 95.44(7)\,a_0$ for $|1,-1\rangle, |2,+1\rangle$ [182]. The data statement reads "All data generated or analysed during this study are included in this published article." [1].

8.2 Coherence observables

In-situ density. The images resolve the 4.35 μm pattern; the 395 nm lattice is not stated to be resolved. The two sound-based f_s experiments in 2D gases compared f_s with Leggett bounds computed from measured profiles [11, 22], and the 2D brackets [26] need $n(\mathbf{r})$ on the scale of every modulation. Because $1/n$ is convex, a density averaged over the unresolved lattice still gives a valid but looser upper bound f^+, while f^- loses its guarantee (our remark). In-situ images also give the local compressibility, strongly suppressed in a Mott domain [101].

Table 8.1: Observables for the TBOL problem. "In TBOL?" refers to Meng *et al.* [1] as read for this map; the last column names the calculation that would predict the observable.

Observable	In TBOL?	What it certifies	Template	Calculation to compare
in-situ density	yes, L_M resolved	density modulation; input to Leggett brackets	[11, 22]	GPE ground state; brackets [26]
compressibility	no	Mott (incompressible) or not	[101]	QMC, GPE equation of state
primary TOF peaks	yes, visibility	coherence on the scale a	[29, 99]	TOF image, near field [2]
moiré TOF peaks	yes, $\Omega_R > 0.5\,E_R$	coherence beyond L_M	[13, 21]	same, at finite T [19]
microwave spectra	yes, $V_0 \leq 10\,E_R$	single-particle bands, calibration	—	band structure
sound speed, κ	no	f_s tensor, $c^2 = f_s/(m\kappa)$	[22]	Bogoliubov sound [84, 165]
phase twist	no	f_s tensor, $E(v)$	[11]	twisted GPE [26]
Josephson oscillations	no	inter-domain coupling; f_s	[46, 67]	GPE plasma mode
stirring, shedding	no	superflow below v_c, with a nucleation model	[6, 37]	time-dependent GPE [15]
Bragg spectra	no	gaps, Bogoliubov bands	[61, 62]	Bogoliubov–de Gennes
modulation, Higgs	no	MI gap; amplitude mode near the MI	[24, 49]	t-DMRG, QMC [127, 151]
winding numbers	theory only	f_s	—	QMC [23, 170]
temperature	not stated	needed by every row	[64, 101]	—

Primary and moiré peaks. The moiré peaks sit at $1/11$ of the primary spacing (since $L_M = 11a$); their visibilities fix phase coherence on the scales a and L_M and nothing more (Chapter 6). Whether 18 ms reaches the far field cannot be judged without the radial size; for a 400 nm lattice with sample size and coherence length near 30 sites, $t_{FF} \approx 100$ ms by the authors' estimate [2]. The quasicrystal criterion reads sharp peaks as long-range coherence and a broad distribution as localised states [13]; applied here, SF-II has coherence over the primary lattice and none detected across moiré cells. A Talbot measurement of the coherence length, as in a disordered 2D lattice where it fell from 15(3) to 1.4(3) sites [21], would turn this into a length.

8.3 Response observables and what each would add

Sound and phase twist. The moiré period lies between the two modulations on which f_s was measured through sound, $3.93(4)\,\mu$m [11] and $6.0(1)\,\mu$m [22]. The template is the triangular lattice: a box of $L = 42(1)\,\mu$m, $\rho_0 = 51(2)\,\mu\text{m}^{-2}$, $g\rho_0/k_B = 45(2)$ nK, $T < 20$ nK and $\omega_z/2\pi = 4.1(2)$ kHz, with the tensor obtained statically, from the density profile and the continuity equation, and dynamically, from the compressibility and the sound speed; the tensor was "nearly isotropic" [22]. For a moiré pattern that keeps the fourfold symmetry of the square lattices, the tensor is isotropic by symmetry (our remark), so the ratio c_x^2/c_y^2 [11] carries no information and the dynamic route transfers. Unlike it, the TBOL gas has two coupled components, a 395 nm lattice under the moiré pattern and a four times lower ω_z. A phase twist of the two-component GPE, $E(v) = E(0) + f_s N m v^2/2$, gives the same tensor in a calculation [26].

Stirring and vortex shedding. A repulsive beam swept through an oblate condensate nucleates vortex pairs above v_c: v_c/c_s is "about 0.4" for the stiffest obstacle, with a dip at $V_0 \approx \mu$ where the obstacle turns from penetrable to impenetrable [16]. In a von Kármán street $v_c = 1.11(5)$ mm/s against $c_s \approx 5.1$ mm/s for an obstacle about 29ξ wide, with regular shedding just above v_c and

a Strouhal number fD/v (shedding frequency f, obstacle width D) saturating near 0.2 [37]. In a 2D gas stirred with a beam of waist 2.0(5) μm, a non-zero v_c appears above a local degeneracy $\mu_{\mathrm{loc}}/k_BT \approx 0.24$ [6]. Against $L_M = 4.35\,\mu$m, a 2 μm beam would stir inside one moiré cell and a 10 μm beam across about two (our remark), testing intra- and inter-domain flow.

Bragg and modulation spectroscopy. Momentum-resolved Bragg spectroscopy maps the bands of a lattice superfluid and shows where Bogoliubov theory applies and where it fails [62]; the Bragg response "drastically drops" in the insulating regime [61]. Modulation spectroscopy shows a broad continuum in the superfluid and a discrete spectrum in the MI, with the second peak at (1.91 ± 0.04) times the first [49]; time-dependent density-matrix renormalisation (t-DMRG) reproduces the peak positions "provided we assume the presence of an incommensurately filled region" [127]. Near the two-dimensional MI, a 3% depth modulation revealed an amplitude (Higgs) mode whose onset softens towards $j_c \simeq 0.06$ [24], a resonance that QMC finds only close to the critical point [151]. In the TBOL these templates would test the "gapped insulator" around the SF-II domains and ask whether the domains carry an amplitude mode. They would also resolve the Bogoliubov bands of the moiré cell of a coherently coupled mixture, in which a finite coupling turns first-order demixing into a second-order transition [172].

The moiré Bose–Hubbard mapping. In 2D the square-lattice benchmark is $(J/U)_c = 0.05974(3)$ [93], $U_c/J = 16.7424(5)$ [94] from QMC, against $5.8z$ in mean field [88]. The model has limits: in hexagonal potentials "the Bose–Hubbard approximation fails to describe but the first Mott lobe" at 15 E_R [104], and multi-body interactions shift the spectrum [91]. For twisted bilayers, Paul, Recher and Santos merge strongly coupled site pairs into clusters and find "mobility islands" [183]; incommensurate twists act as effective disorder with multifractal states [184]; without interlayer tunnelling, contact interactions give interlocked Mott phases [185]. No such benchmark was found for single-site Gutzwiller in the moiré case.

Josephson array or Bose glass. If SF-II is an array of coherent domains coupled by tunnelling, its f_s is set by the inter-domain band mass: in the modulated gas f_s "exactly coincides with the ratio m/m_x^*" [11], and the sound speed follows from the Taylor–Zaremba expansion of the Bogoliubov equations [85]; an array oscillates coherently at small amplitude [46], and single cells give f_s [67]. A Bose glass is instead "an insulating but compressible state without long-range phase coherence" [13]: QMC finds $f_s \approx 0$ there while the compressibility changes smoothly [23], a disorder quench found evidence for "superfluid puddles in the Bose glass phase" [147], and in a twisted lattice model stronger interactions turn a disconnected glass into a connected superfluid at low filling [186]. Separating the readings takes compressibility and f_s together.

Temperature and calculations. Freezing the axial motion needs $\hbar\omega_z$ well above k_BT and gn [31, 32]; the TBOL pancake has $\omega_z \approx 2\pi \times 1$ kHz, against 1.9(2) kHz in the equation-of-state experiment of Yefsah *et al.* [41], but without T and n the two ratios cannot be formed. For thin domains temperature decides: QMC of a single-component twisted potential at $k_BT = 0.01$–$0.04\,E_R$ finds maximal hoppings of 10^{-11}–$10^{-3}\,E_R$ in the superfluid domains and "thermal broadening strongly depletes the SF domains" [170]. Simulations must assume these inputs, e.g. $N = 2 \times 10^4$ and a trap of $2\pi \times (20, 20, 1000)$ Hz in a Floquet GPE study [187] (preprint). A uniform Ω_R maps onto a scalar moiré potential of amplitude proportional to Ω_R [188] (preprint) and, at large detuning, onto Hofstadter's model plus long-range hopping [189]; Kariyado–Vishwanath flat bands need a spatially varying coupling [190].

Key points

- TBOL measured coherence, the in-situ moiré density and microwave spectra; tempera-

ture, radial trap frequency and pancake atom number were not found in the paper.

- The f_s template is sound in a triangular lattice of period 6.0(1) μm.
- Stirring and modulation are unused templates; Josephson and compressibility data would separate a domain array from a glass.

9 Open questions

Twenty-two questions in five themes, each with its evidence, and directions no checked study took up.

The questions follow from Chapters 6 and 8 (Figure 9.1); "found" refers to the checked literature.

definitions, Landau	**Q1** Landau or instability	**Q2** v_c/c_s in flat geometries	**Q3** which f_s at $T > 0$	**Q4** brackets in disorder	
measuring f_s	**Q5** deep lattices	**Q6** sound near BKT	**Q7** sound in disorder	**Q8** Josephson route	**Q9** new probes
Bose–Hubbard observables	**Q10** visibility against f_s	**Q11** critical points in traps	**Q12** beyond one band	**Q13** Higgs in disorder	**Q14** methods at finite T
disorder, localisation	**Q15** Talbot length vs f_s	**Q16** Bragg broadening	**Q17** mobility edge	**Q18** quasicrystals, $T > 0$	
twisted bilayer	**Q19** is SF-II superfluid?	**Q20** effective model	**Q21** extended condensate	**Q22** T, two components	

Figure 9.1: The 22 open questions, one row per theme; the rows follow the subsections below.

9.1 Definitions and the Landau criterion

Q1 **Landau or dynamical instability.** Burger *et al.* read the breakdown in a moving lattice as the emission of elementary excitations at the local sound speed, naming neither the Landau criterion nor an instability [45]. Wu and Niu suggested a dynamical instability [17]; later moving-lattice experiments identified dynamical instability [47] and two-atom scattering [80] in their own data. The two instabilities are distinct [17, 78] and were seen in one experiment [18]; in 1D, damping sets in, attributed to quantum fluctuations [79], below the tight-binding threshold $|q| = \pi/(2a)$ [73]. Which criterion sets v_c in a 2D or moiré lattice?

Q2 **Why stirring gives higher v_c/c_s in flat geometries.** Stirring gives 0.3–0.6 in a 2D gas [6], "about 0.4" in an oblate gas [16] and ≈0.6 in a ring [38], against about a quarter [5] and 0.07–0.11 [14] in 3D; yet oblate samples give ∼0.1 [36] and 1.11(5) against 5.1 mm/s [37]. Penetrability [16], low-density edges [76], obstacle size (as reported [73]) and non-equilibrium [77] enter. A size series at fixed dimension would separate the two.

Q3 **Which f_s at $T > 0$.** The drag response [26], windings [83], the BKT jump $n_s\lambda^2 = 4$ [32], or $\Delta\rho_s = 2mT_{KT}/\pi$ in a trap [110], and momentum fluctuations of a spin-1 gas [191] are distinct routes to a tensor [81]; no comparison for one modulated 2D gas was found.

Q4 **Tight brackets in 2D disorder.** The bound is an identity in 1D [11], holds in a triangular lattice [22], is "tighter for a 2D optical potential than for a grid of columns" [26] and depends on angle [87] (preprint); for 2D speckle it is "too loose of an estimate" [165]. Tight brackets for 2D disorder or two length scales are missing.

9.2 Measuring the superfluid fraction in lattices and two dimensions

Q5 **Deep lattices.** Sound reached $V_0 \simeq 4.7\,\mu_0$ [22]; Bragg resonances follow Bogoliubov bands for $s < 6$ [48], with $c = 1/\sqrt{\kappa m^*}$ [84]; deeper, they fall faster than predicted [63], depletion exceeds 50% [103], the Mott spectrum is discrete [49]. Does sound give f_s there?

Q6 **Sound near BKT with a lattice.** In a box $c = 1.49(3)$ against a Bogoliubov $1.6(1)$ mm/s, as reported [44], fitted by Bogoliubov theory with the finite-T superfluid density [132]; c_2/c_0 is predicted to fall from 0.56 to 0 at $\tilde{g} = 0.1$ [138], and D_s jumps from 0 to 4 [8]. A lattice added to the box would show how a modulation changes each of them.

Q7 **Sound in disorder.** In speckle $mc^2 = f_s(n\kappa)^{-1}$; effects should be observable "in current experiments, similarly to the measurement of the speed of sound in periodic potentials" [165] (Bogoliubov excitations in disorder: [192]); no measurement was found.

Q8 **The Josephson route.** Supersolid cells give f_s [67]; a lattice array oscillates coherently at small amplitude [46]; coupled 1D gases realise the sine-Gordon model, as reported [174]. Can the same route give f_s in a 2D optical lattice?

Q9 **New probes.** Microscopes detect atoms on single sites with probability 94.9(7)% [50]; noise correlations [121] and Edwards–Anderson parameters [21] add order. Each records order, not a response; a bound on the stiffness would need a model.

9.3 Bose–Hubbard observables

Q10 **Visibility against f_s.** Condensate fractions (≈45%, zero at $U/J = 15.8(20)$ in 2D [130]), parameter-free momentum distributions [35], Mott visibilities [99, 100] and number squeezing [102] are measured; peaks survive above T_c [19]; no visibility–f_s relation at experimental T was found.

Q11 **Critical points in traps.** The 3D $(U/J)_c = 34.2(2.0)$ lies 2σ from QMC $29.34(2)$ [80, 92]; in 2D QMC gives $(J/U)_c = 0.05974(3)$ [93], $U_c/J = 16.7424(5)$ [94] against a mean-field 23.31 [89, 90], in 1D $(U/J)_c \approx 3.61$ (our inverse of $J_c/U = 0.277(10)$) [95]. What remains is to set the f_s onset against the coherence onset at $T > 0$ [133, 153].

Q12 **Beyond one band.** Multi-body terms shift the U and $2U$ resonances [91], the model fails beyond the first Mott lobe in hexagonal potentials [104], and a resonance above the first Bogoliubov band was seen [148]. Which observable is most sensitive to these corrections?

Q13 **Higgs mode in disorder.** The amplitude mode softens towards $j_c \simeq 0.06$ [24], pairs with a Goldstone mode [149], carries an infrared singularity from Goldstone contamination [152], and is no resonance at 20% detuning [151]; a 1D Bose glass shows an extra peak in the Mott gap, as reported [66]. No 2D measurement was found.

Q14 **Methods at $T > 0$.** Ramps give an exponent 0.31 against the Kibble–Zurek 0.75, as reported [158], and exponents that "strongly depend on the final interaction strength" [154]. Worm QMC handles arbitrary lattice structures [193]; truncated Wigner is "not suitable" in the strongly interacting limit [134], and a cluster variant uses 2×2 clusters [135] (preprint). Which of them gives f_s of a modulated 2D gas?

9.4 Disorder and localised excitations

Q15 **Talbot length against f_s.** QMC finds $f_s \approx 0$ in the quasicrystal glass, except close to the separation line [23], and in 1D, as reported, $f_s = 0$ with sizable compressibility [166]; the

glass always separates SF and MI [94, 146]. It is seen through coherence [13], a Talbot length of 1.4(3) sites, which the authors attribute to the Bose-glass phase [21], transport [54] and quenches [147]. Does that length track f_s?

Q16 **Bragg broadening.** Localised Bogoliubov quasiparticles are screened at low momentum; "We expect that the AL of BQPs could be observed directly, for instance as a broadening of the resonance lines in Bragg spectroscopy" [162], with a Lifshits glass among the ground states, as reported [144]. Flat spectra were reported [65, 159]; the broadening was not.

Q17 **Interactions and the mobility edge.** In 3D, $E_c = 52(5)$ nK at $V_R = 46.8(1.4)$ nK lies 20–40% above theory [52]; in 1D, localisation [9, 10] gives way under interactions to "locally coherent fragments" [53] and subdiffusion [157]. Their effect on f_s is unmeasured.

Q18 **Quasicrystals at $T > 0$.** The glass cannot be traversed adiabatically on experimental timescales [13]; quenches into it give a momentum-space quantum walk [57] (preprint); QMC finds it up to $T \simeq 0.5$–$8\, E_R/k_B$ depending on interaction [167]. For the eightfold lattice [55] no response measurement was found.

9.5 The twisted-bilayer target

Q19 **Is SF-II superfluid?** "Superfluid only with short-range coherence" is a coherence label [1]. Theory finds "mobility islands" [183], interlocked Mott phases [185] and thermally depleted domains [170]; a compressibility or sound measurement would tell them apart.

Q20 **Which effective model.** The scalar potential [188] (preprint), the Hofstadter limit [189], the cluster model [183] and the effective disorder of incommensurate twists [184] are not joined, and a uniform coupling gives no Kariyado–Vishwanath flat band [190].

Q21 **The extended condensate.** A quench of a spin-1 condensate in a twisted lattice leaves vortex pairs [194] (preprint), and a Floquet study modulates the interaction strength in time [187] (preprint); neither gives Bogoliubov spectra.

Q22 **Temperature and two components.** Coupled 1D gases give a junction limit [175, 176]. At what temperature do thin domains deplete [170] with two coupled components, and how do demixing [172], miscibility [182] and false-vacuum decay [173] enter a moiré cell?

9.6 Directions not yet systematically addressed

As far as the checks for this map could establish, no study has taken up the following.

D1 An f_s measurement on a moiré or Rabi-coupled gas (so far: [11, 22, 67]).

D2 A direct f_s, sound or Bragg measurement of a Bose glass, as proposed in [162, 165].

D3 Visibility and f_s computed for one trapped, modulated 2D gas at the expansion time [2].

D4 Bogoliubov spectra, sound and f_s tensor of the Rabi-coupled twisted condensate.

D5 Ramps or quenches of the coupling (other quenches: [186, 187, 194]).

D6 Leggett brackets from the measured density of a potential with two length scales.

D7 Amplitude-mode spectroscopy, or stirring, in a disordered, quasi-periodic or moiré lattice.

Key points

- What sets v_c in a lattice, and why v_c/c_s spans 0.07–0.6, is not settled.
- No f_s measurement on a moiré or disordered gas, and no f_s–visibility tie, was found.
- For the twisted bilayer, a response measurement and an effective model are both missing.

10 Reading paths and outlook

Where to start, by background, and what the next measurements and calculations would settle.

Where to start. Theorists new to the laboratory should read chapter 3, then the 2D review of Hadzibabic and Dalibard [32], the review of Bloch, Dalibard and Zwerger [97], Navon *et al.* on homogeneous gases [137], and the lecture notes of Dalibard (coherence, Landau's criterion, lattice instabilities) [73] and of Perrin on two-dimensional gases (preprint) [116]. What an image shows is set by the far-field analysis of Gerbier *et al.* [2] and by the QMC finding that sharp peaks survive above T_c [19]; Pitaevskii and Stringari's monograph is the reference [195].

For the critical velocity, read the stirring experiments of Raman *et al.* and Onofrio *et al.* [5, 14] next to the simulations of Winiecki *et al.* (cylinder) and Jackson *et al.* (trapped gas) [15, 76], then the vortex-nucleation work of Kwon *et al.* [16, 37]; Astrakharchik and Pitaevskii give the impurity drag in 1D to 3D [74]. For lattices, Wu and Niu define energetic and dynamical instabilities [17], and De Sarlo *et al.* measured both [18]. For definitions, start from Leggett [25, 72, 169], then Pérez-Cruz *et al.* (drag, 2D brackets) [26] and Chauveau *et al.* (sound) [11].

Numerical analysts: models in chapter 2, checks in chapter 4. Gross–Pitaevskii and Bogoliubov calculations [68, 69] are tested against the Bragg data [3, 4] and the lattice Bogoliubov bands and sound speeds [71, 84, 85]. Bose–Hubbard codes are tested against the mean-field tip of van Oosten *et al.* [89], the QMC and DMRG critical points [92, 93, 95] and the cluster-Gutzwiller extrapolation [196]; an open worm-algorithm code exists [193], and Krutitsky reviews lattice results [98]. 2D simulations are tested against the classical-field constants [107, 108] and the trapped QMC of Holzmann and Krauth [110]; for the validity of classical-field and truncated-Wigner methods see Sinatra *et al.*, Blakie *et al.*, Polkovnikov and Rooney *et al.* [197–200].

Outlook. The superfluid fraction of a modulated condensate has been measured through sound in two laboratories [11, 12, 22] (Tao *et al.*: sound, with f_s via BdG theory) and through Josephson oscillations in a third [67], all in condensates away from an insulator. The observables of this map can disagree further out: near the Mott transition, where the critical temperature is suppressed [153]; in the Bose glass, where QMC gives $f_s \approx 0$ [23] and no measurement of f_s was found for this map; near the BKT point, where the measured coherence exponent departs from its universal value [42]; and in moving lattices, where energetic and dynamical instabilities compete. A response measurement in any of these, against a calculation of the same quantity, would test the dictionary that chapter 8 applies to the twisted bilayer.

Key points

- Theorists: the toolbox chapter, the reviews and lecture notes, then what an image shows.
- Critical velocity: stirring experiments, vortex-nucleation simulations, Wu and Niu.
- Definitions: Leggett, Pérez-Cruz *et al.*, Chauveau *et al.*; numerics: Bragg, QMC, BKT.

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A Glossary

Terms from superfluidity, two-dimensional gases, optical lattices, disorder and cold-atom measurement, as this map uses them.

Landau criterion. A fluid flowing past a wall or obstacle at velocity v can create an excitation of momentum p and energy $\varepsilon(p)$ only if $v > \varepsilon(p)/p$; below the minimum of this ratio the flow cannot decay by emitting excitations. A necessary condition for metastable superflow.

Critical velocity. The velocity v_c above which a moving obstacle or a flow starts to dissipate. For a Bogoliubov spectrum Landau's value is the sound speed; measured values lie well below it, and vortex nucleation is one reason.

Strouhal number. $\mathrm{St} = fD/v$, the vortex-shedding frequency f times the obstacle width D divided by the flow velocity v.

Superfluid fraction. The fraction f_s of the mass that does not follow a slow motion of the walls or of the potential. In an anisotropic potential it is a tensor.

Non-classical rotational or translational inertia. The reduction of the moment of inertia (or of the translational mass) below its classical value, the operational definition of f_s in a rotating or moving frame.

Leggett bound. An upper bound on f_s computed from the density profile alone; for a weakly interacting condensate in a one-dimensional modulation it is an equality. In two dimensions upper and lower brackets exist.

Phase twist. Imposing a phase gradient on the order parameter, by twisted boundary conditions or a moving frame; the energy cost at small twist is proportional to f_s. Called the helicity modulus in lattice models.

Winding number. In path-integral quantum Monte Carlo, the number of times world lines wrap around the periodic box; its variance gives f_s.

Bogoliubov spectrum. The excitation spectrum of a weakly interacting condensate, $\varepsilon(k) = \sqrt{\epsilon_k(\epsilon_k + 2gn)}$ with $\epsilon_k = \hbar^2k^2/2m$: phonons with speed $c_s = \sqrt{gn/m}$ at small k, free particles at large k.

Healing length. $\xi = \hbar/\sqrt{2mgn}$ (conventions vary by $\sqrt{2}$): the length over which a condensate recovers from a local perturbation; the crossover scale of the Bogoliubov spectrum.

Quasi-condensate. A gas whose density fluctuations are suppressed while its phase still fluctuates over large distances; it looks bimodal in an image but need not be superfluid.

BKT transition. Berezinskii–Kosterlitz–Thouless transition: the superfluid transition of a two-dimensional gas, driven by the unbinding of vortex pairs; below it coherence decays algebraically and the superfluid density jumps to $n_s\lambda^2 = 4$ at T_c.

Phase-space density. $D = n\lambda^2$ with λ the thermal de Broglie wavelength; the BKT point is at $D_c = \ln(C/\tilde{g})$.

Two-dimensional coupling. The dimensionless constant $\tilde{g} = \sqrt{8\pi}\, a_s/a_z$, with a_s the scattering length and a_z the oscillator length of the tight axis.

Quasi-2D. The regime where the tight-axis energy $\hbar\omega_z$ exceeds both k_BT and the interaction energy, so that the motion along z is frozen while collisions remain three-dimensional.

Accordion lattice. A standing wave made by two beams crossing at a variable angle; changing the angle changes the spacing and compresses a single plane of atoms.

Rabi coupling. A microwave or Raman field that coherently couples two internal states, with Rabi frequency Ω_R; in the twisted-bilayer lattice it couples the two layers.

Box trap. A potential that is flat inside and steep at the walls, made with shaped light (an intensity mask or a digital micromirror device, DMD); it gives a uniform gas.

Time of flight (TOF). Imaging after the trap is switched off and the cloud has expanded; in the far field the image reflects the in-trap momentum distribution.

Far field. The expansion time after which the initial size of the cloud no longer matters; for a coherent cloud it can be much longer than the flight times used.

Band mapping. Switching a lattice off slowly compared with the band gap but fast compared with tunnelling, so that each quasimomentum maps onto a free momentum and each band onto its Brillouin zone.

Bragg spectroscopy. Two laser beams with momentum difference $\hbar\mathbf{q}$ and frequency difference ω transfer atoms; the transfer measures the dynamic structure factor $S(\mathbf{q}, \omega)$.

Lattice-modulation spectroscopy. Periodic modulation of the lattice depth; the energy absorbed at each frequency reveals gaps and excitation bands.

Amplitude (Higgs) mode. The oscillation of the magnitude of the order parameter near the superfluid–Mott-insulator transition, whose gap softens at the critical point.

Bose–Hubbard model. Lattice model of bosons with nearest-neighbour hopping J (written t in several cited papers) and on-site repulsion U; valid for a deep lattice with a well-separated lowest band. In 2D the superfluid–Mott transition at unit filling lies at $j_c = (J/U)_c \simeq 0.06$.

Gutzwiller ansatz. A product over lattice sites of local states, each expanded in the Fock basis: the site-decoupled mean-field treatment of the Bose–Hubbard model. Cluster variants use products of multi-site clusters.

Mott insulator. The Bose–Hubbard phase at large U/J and integer filling: fixed site occupations, a gap and no long-range phase coherence.

Bose glass. An insulating but compressible, gapless phase of bosons in disorder, without long-range phase coherence.

Edwards–Anderson parameter. A glass order parameter q: it measures how strongly a local quantity is frozen into a random, site-dependent pattern and vanishes in a homogeneous phase. In a site-resolved experiment on a disordered lattice $q \approx 0.06$, against a possible maximum of 0.25, in a regime which the authors attribute to the Bose-glass phase [21].

Lifshits glass. The state of weakly interacting bosons in disorder that occupy separate localised states of the low-energy (Lifshits) tail of the spectrum; stronger interactions turn it into a fragmented and then a delocalised disordered condensate [144].

Anderson localisation. Exponential localisation of non-interacting waves by interference of multiple scattering in a disordered potential.

Mobility edge. The energy that separates localised from extended single-particle states in a disordered potential (in three dimensions).

Speckle. The random intensity pattern of laser light passed through a diffuser; a disorder potential with a finite correlation length.

Bichromatic lattice. Two superposed lattices with incommensurate periods; a quasi-periodic potential that realises the Aubry–André model in one dimension.

Optical quasicrystal. A potential made of several lattices whose angles give long-range order but no periodicity, such as the eightfold lattice of four one-dimensional lattices at 45°.

Quantum-gas microscope. High-resolution fluorescence imaging of a lattice gas with single-site resolution; it detects the occupation of each site modulo two.

Josephson array. A chain or lattice of condensates coupled by tunnelling; the frequency of small oscillations gives the Josephson coupling, and in a supersolid the superfluid fraction.